

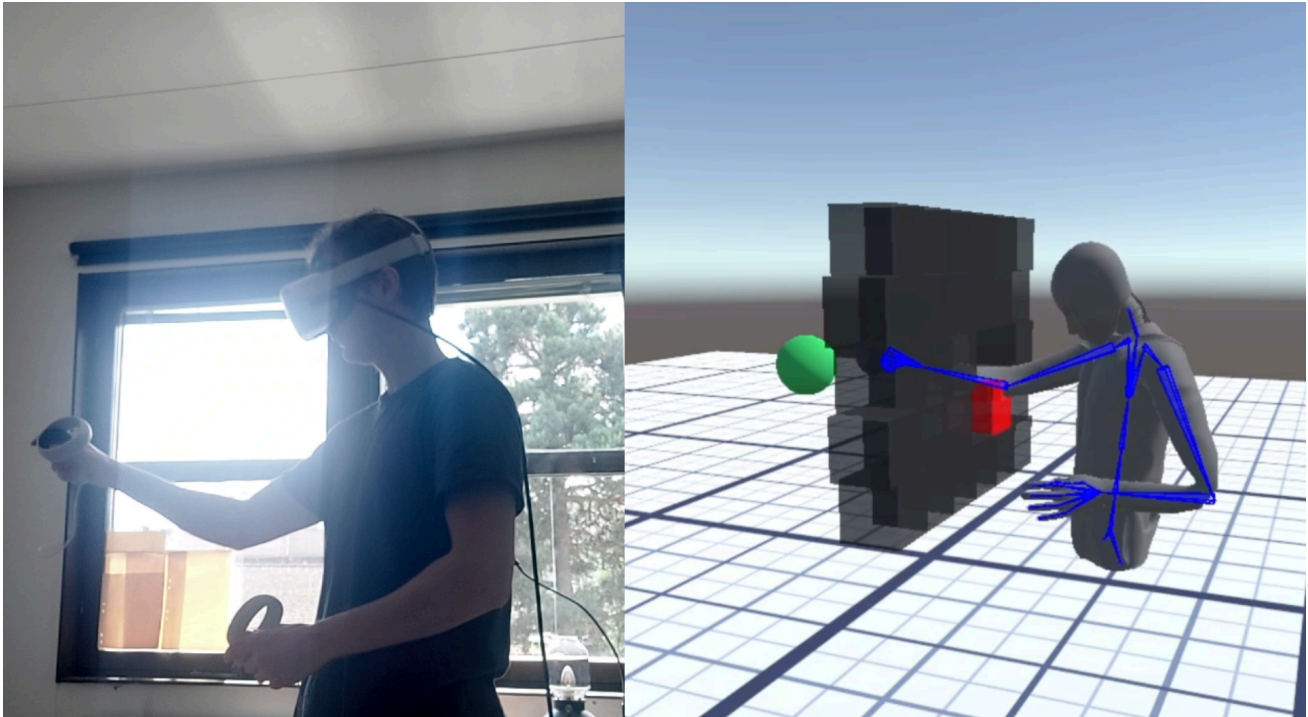


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# Disturbed Self-Avatar Motion Synchrony Severely Impacts the Sense of Agency in VR Manipulation Task

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## Abstract

Self-avatars in VR applications have seen increased use in recent years due to VR hardware becoming more accessible. The feeling of being in control of the self-avatar can be measured via the sense of agency. To induce a sense of agency it is essential that there is motion synchrony, both temporal and positional, between the avatar’s virtual limbs and the user’s real limbs. Disturbed synchrony, especially via motion delay, causes a mismatch in sensorimotor feedback, and has been shown to disrupt this feeling. Self-avatar motion delay has been investigated in various scenarios and at different delay conditions. However, in studies where the avatar is shown from a first-person point of view - mainly using motion tasks that do not involve interacting with the virtual environment and at limited delay conditions. Furthermore, exploration of similar effects from offsets in limb position has seen little interest. In this paper, we study the individual impact of motion delay (milliseconds) {B, D98: 98 ms, D165: 165 ms, D243: 243 ms} and offset (centimeters) {B, O06: 6 cm, O12: 12 cm, O20: 20 cm} as well as combinations of both {B, M1: (98 ms, 12 cm), M2: (165 ms, 6 cm), M3: (243 ms, 20 cm)} on the user’s ( $n = 22$ ) self-avatar arms in a manipulation task that focus on arm motion. Our baseline condition B has 32 ms delay and less than 1 cm offset. The self-avatar is of the upper body and uses inverse kinematics to animate the arms, with the head and hands targeted to the head-mounted display and controllers of a Quest 2 VR headset. We evaluate the motion synchrony conditions qualitatively (sense of agency and sense of body ownership) and quantitatively (task performance metrics), as well as try to gain insight into how users perceive the disturbed motion synchrony via pre and post-questionnaires. Our results show that both delay and offset affect the sense of agency, sense of body ownership, and task performance. With our conditions, delay has a larger impact on the evaluation metrics than offset, with a significant negative effect on the sense of agency even between 32 ms and 98 ms. Surprisingly and contrary to previous research, the sense of agency is impacted by smaller disruptions in synchrony compared to the performance metrics. We speculate this could be because user performance in our task is not as sensitive to disturbed motion synchrony. We suggest that when embodying a self-avatar and using it to perform tasks in the virtual environment where one has to be highly aware of their body motion, synchrony between the user’s and the self-avatar’s motion is critical.

**Keywords:** motion synchrony; delay; offset; sense of agency; motor task; arms; self-avatar; virtual reality

# Störd rörelsesynkronisering i självavatarer påverkar allvarligt känslan av rörelsekontroll i VR-manipuleringsuppgift

## Sammanfattning

Självavatarer i VR-applikationer har fått ökad användning under de senaste åren i takt med att VR hårdvara blivit mer tillgänglig. Känslan av att ha kontroll över sin egen avatar kan mätas via sense of agency (ung. känslan av rörelsekontroll). För att framkalla sense of agency är det viktigt att det finns en rörelsesynkronisering, både temporal och positionell, mellan mellan avatarrens virtuella lemmar och användarens verkliga lemmar. Störd synkronisering, särskilt via rörelsefördröjning, orsakar en mismatchning i sensorimotorisk feedback och har visat sig störa den här känslan. Rörelsefördröjning hos självavatarer har undersökts i olika scenarier och vid olika fördröjningsförhållanden. Men i studier där avataren visas från en första persons synvinkel - främst med hjälp av rörelseuppgifter som inte innebär interaktion med den virtuella miljön, och vid begränsade fördröjningsförhållanden. Dessutom har utforskningen av liknande effekter från förskjutningar i lemmposition har inte varit av något större intresse. I detta dokument studerar vi den individuella effekten av rörelsefördröjning (millisekunder) {B, D98: 98 ms, D165: 165 ms, D243: 243 ms} och förskjutning (centimeter) {B, O06: 6 cm, O12: 12 cm, O20: 20 cm} samt av kombinationer av båda {B, M1: (98 ms, 12 cm), M2: (165 ms, 6 cm), M3: (243 ms, 20 cm)} på användarnas ( $n = 22$ ) självavatararmar i en manipuleringsuppgift som fokuserar på armrörelser. Grundtillståndet B har 32 ms fördröjning och mindre än 1 cm förskjutning. Självavataren består av en överkropp och använder inverterad kinematik för att animera armarna, med huvud och händer riktade mot den huvudmonterad skärmen och kontrollerna i ett Quest 2 VR-headset. Vi utvärderar rörelsesynkronitetsförhållandena kvalitativt (sense of agency och sense of body ownership) och kvantitativt (mätvärden för uppgiftsprestation), samt försöker få insikt i hur användarna uppfattar den störda rörelsesynkroniseringen via frågeformulär före och efter. Våra resultat visar att både fördröjning och förskjutning påverkar sense of agency, sense of body ownership (ung. känslan av kroppsägarande) och uppgiftsprestation. Med våra förhållanden har fördröjning en större inverkan på utvärderingsmått än förskjutning, med en signifikant negativ effekt på sense of agency även mellan 32 ms och 98 ms. Överraskande nog och i motsats till tidigare forskning påverkas de kvalitativa mätvärdena vid mindre störningar i synkroniseringen jämfört med prestationsmått. Vi spekulerar i att detta kan bero på att användarnas prestation i vår uppgift inte är lika känslig för störd rörelsesynkronisering. Vi föreslår att när man förkroppsligar en självavatar och använder den för att utföra uppgifter i den virtuella miljön där användaren måste vara mycket medveten om sin kroppsrorelse, är synkronisering mellan användarens och självavatarrens rörelse kritisk.

**Nyckelord:** rörelsesynkroni; fördröjning; förskjutning; känsla av rörelsekontroll; motorik uppgift; armar; självavatar; virtuell verklighet

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# 1 Introduction

Immersive virtual reality (VR) using head-mounted displays (HMDs) and motion controllers enables us to nearly fully immerse ourselves within virtual environments. Feeling present in the virtual environment is a key feature of VR experiences, and an essential part contributing to this feeling is having a virtual representation of yourself. Virtual humanoid self-avatars that are centered at the user’s position and match the user’s limb motion have been shown to be important in this regard [39, 15, 11]. Synchrony, both temporal and positional, between the avatar’s virtual limbs and the user’s real limbs, is critical to not cause a mismatch in sensorimotor feedback. The subjective feeling of being in control of a self-avatar’s motion and actions can be measured via the sense of agency (SoA) and is central in VR self-avatar research. The SoA is the subjective feeling of being in control of your actions and external events through them. Studying the SoA is a subfield of its own, and a lot of factors have been found to affect it, such as task performance, mental state, and social interactions [49]. Much work has been put into investigating how different disturbed synchrony conditions in self-avatars, especially delay, affect the SoA. Measuring task performance together with SoA is also of interest. Kiltner et al. [27] suggests that a high SoA should correlate with high task performance under the assumption that a more finely controlled self-avatar results in better task performance. The present paper focuses on the effects of disrupted motion synchrony via delay and offset in self-avatar motion.

Several related studies on self-avatars have investigated motion delay in particular. Koiliadis et al. [28] explored the effect of several tracking artifacts on self-avatars from different viewing perspectives (first-person view of the avatar, viewing the avatar in a mirror, view while walking). Their single delay condition of 150 ms (milliseconds) was found to significantly reduce the SoA in all perspectives except while walking. Waltemate et al. [46] used a mirror viewing perspective of the self-avatar (non-HMD) and compared several delay conditions. They found a significant reduction in the SoA and sense of body ownership (SoBO, approx. the feeling that the body belongs to you), between 45 ms and 125 ms. They also found that task performance was affected between 45 ms and 75 ms, and that task performance predicted SoA and SoBO in their task. The task involved tracing the motion of pre-recorded animations. The results from these studies suggest that delay can have an effect at relatively low values, but that it depends on both the task and the viewing perspective of the self-avatar. Waltemate et al. suggest that harder motor tasks where the user is actively considering the position of body parts involved in the task will be able to detect the effects of lower delays.

Position offsets on joints on the self-avatar are not nearly as widely studied as delay. Static offset was studied in a 1998 paper by Groen and Werkhoven [18], who investigated the impact of a 10 cm (centimeters) lateral offset on users’ hand position while immersed in a virtual environment (non-HMD) on the users’ ability to adapt their hand-eye coordination. They found no decrease in manipulation performance with the offset applied. Bovet et al. [6] found

offsets of 7.5 cm to negatively impact the SoA over a self-avatar when self-contact was involved. Aoyagi et al. [1] had users perform a tracing task where a sphere was mapped to their VR controller. They found that when the sphere was offset in a way that made it appear to the user like they performed the task better, they felt a significantly higher SoA over the sphere. Based on this, the contrary might be true. If offset is applied in a way that makes perceived task performance worse it might affect the SoA negatively.

There have also been self-avatar studies that have had problems with motion tracking and report delays or offsets as factors affecting their results. For instance, when comparing three different self-avatar conditions, one with superior pose accuracy (motion capture), to another with lower pose accuracy but lower delay and less tracker drift (sparse tracking with estimation of missing points), Yun et al. [52] found that the non-motion capture condition sometimes performed as well or better concerning task performance and SoA. The authors suggest this was because of the difference in delay (33 ms vs 91 ms) and tracker drift. However, it is hard to know since no one has investigated such low delay values when viewing a self-avatar from a first-person perspective, and in motor performance-intensive tasks such as those used in their study. A similar result was reported in [34].

Based on our best knowledge, no study on VR self-avatars where the avatar is viewed from a first-person perspective (the most common perspective in VR experiences) utilizes tasks where the user has to interact with the virtual environment, to measure the effects of either delay or offset, on the SoA and task performance. Interactions with the virtual environment are common in VR applications. Therefore, it is important to understand how the SoA is impacted in such scenarios to be able to increase immersiveness and promote user interaction in VR applications. In this paper, we study the individual effects of motion delay and arm offset in VR self-avatars, as well as combinations of them, on mainly the SoA (qualitatively) and task performance (quantitatively) in a manipulation task. While a smaller focus of study, the SoBO is also measured (qualitatively) for comparison with [46] and [28]. We build a prototype application using Unity3D containing an upper body self-avatar, functionality to apply delay and offset, and a user study scenario with controls and data collection. The upper body self-avatar has hands, arms, a torso, and a head, where the head and hands are mapped to a Meta Quest 2 VR headset and controllers. We use inverse kinematics to estimate the elbow positions and a simple algorithm for orienting the torso (which also determines the shoulder positions). The user study experiment uses a repeated measures design. Participants perform tasks where they pick up objects with their virtual hands and navigate them to target positions while avoiding collisions with moving obstacles. A similar task was used in [52] to measure SoA and task performance for the upper body since this type of collision task makes participants aware of their arm and hand positions [9]. In addition to the mentioned qualitative and quantitative measures, we try to gain some insight into how participants attribute SoA under the presence of different synchrony conditions via a post-questionnaire. As well as explore correlations to immersive tendencies via a pre-questionnaire.

## 1.1 Research questions

**General RQ (GRQ):** How does disturbed synchrony between real and virtual arm motions in virtual reality self-avatars affect the sense of agency? Section 4.2.

Given a self-avatar of the upper body (hands, arms, torso, head) animated using inverse kinematics with hands mapped to a pair of VR motion controllers, and head mapped to a VR headset, with controllable offset between the self-avatar hands and the VR controllers as lateral offsets to the avatars body forward orientation, and controllable delay on the self-avatar hands, arms, and torso motion via buffered tracking updates, measured via a task-based user study where the user performs a manipulation task inside the virtual environment:

### Main RQs:

- **MRQ1:** What is the effect of increasing the motion delay {B, D98: 98 ms, D165: 165 ms, D243: 243 ms} on the sense of agency and task performance? Section 4.2.1.
- **MRQ2:** What is the effect of increasing the position offset {B, O06: 6 cm, O12: 12 cm, O20: 20 cm} on the sense of agency and task performance? Section 4.2.2.
- **MRQ3:** What is the effect of combining both delay and offset {B, M1: (98 ms, 12 cm), M2: (165 ms, 6 cm), M3: (243 ms, 20 cm)} on the sense of agency and task performance? Section 4.2.3.

The main research questions aid in answering the general one. Motion delays are in milliseconds (ms) and position offsets are in centimeters (cm). The baseline condition B has 32 ms delay and less than 1 cm offset (Section 3.1). A description of the task can be found in Section 3.2.2. The questionnaire (Q) to measure the sense of agency and sense of body ownership, as well as task performance metrics, can be found in Section 3.2.5.

### Peripheral RQs:

- **PRQ1:** What is the effect of delay and offset on the sense of body ownership? Section 4.2.4.
- **PRQ2:** Is there a relationship between immersive tendencies and the sense of agency and sense of body ownership? If so, how strong is it? Section 4.2.5.
- **PRQ3:** How do users attribute a sense of agency over the self-avatar arms in the presence of disruptions in synchrony while performing the task? Section 4.2.6.

The peripheral research questions have weaker measurement devices which results in a weaker evaluation than for the main research questions, but are included to reason about the SoA in the task (PRQ3), and to compare with previous studies (PRQ1 and PRQ2). The pre-questionnaire on immersive tendencies (sITQ) related to PRQ2 can be found in Section 3.2.5 Table 3.4, and the post-questionnaire (PQ) related to PRQ3 can be found in Section 3.2.5 Table 3.5.

## 1.2 Delimitations

The self-avatar does not appear realistic or similar to the participants in the user study, a generic 3D humanoid model is used. The user study focuses only on upper body motion. Arm motion in particular. The tasks and the scenario is designed accordingly. The inverse kinematics method used to animate the arms of the self-avatar is built using Unity’s own animation rigging package, and does not always produce the most accurate self-avatar poses (e.g. the elbow position is not always accurate to the user’s elbow position, however the tasks are designed with this limitation in mind).

## 2 Background

This section of the report covers background necessary to better understand the methodology, results, and discussion. Specifically, it briefly introduces self-avatars in VR, and describes the sense of agency (SoA) as well as related concepts. We also present related work on how delay and offset has been found to affect the SoA and task performance, both with and without the use of self-avatars in VR. When discussing delays, we use milliseconds (ms), and when discussing offsets we use centimeters (cm). An overview of findings related to delay and offsets effect on the SoA, sense of body ownership (SoBO), and performance in various tasks can be found in Table 2.1.

Table 2.1. Displays a summary of relevant findings from related studies that researched the effect of delay or offset on the sense of agency (SoA) both with and without VR and self-avatars. This is not an exhaustive list. In the view column, "1st" means first-person view of the self-avatar, and "mirror" means it was shown in a mirror. The findings column indicates where the first negative impact was on a metric. The conditions only include delay or offset, even if other conditions were used in the study as well. The metrics only include SoA, sense of body ownership (SoBO) and performance, even if other metrics were used in the listed study as well. Delays are in milliseconds (ms) and offsets are in centimeters (cm). If baseline is known it is marked in parenthesis as B. SoE abbreviates sense of embodiment (row 3).

| Study                           | HMD                  | Self-avatar     | View                           | Task                           | Conditions                                     | Metrics                | Findings                                   |
|---------------------------------|----------------------|-----------------|--------------------------------|--------------------------------|------------------------------------------------|------------------------|--------------------------------------------|
| Roth and Latoschik [36] (2020)  | Yes                  | Yes (full body) | 1st, mirror                    | Instructed movement            | Delay: 90 (B), 206, 353                        | SoA, SoBO              | SoA, SoBO at 353 ms                        |
| Koiliias et al. [28] (2019)     | Yes                  | Yes (full body) | 1st, mirror, 1st while walking | Observe your own movements     | Delay: 150, Offset: 20 (popping)               | SoA, SoBO              | No effect on SoBO. SoA at both conditions. |
| Toothman and Neff [43] (2019)   | Yes                  | Yes (full body) | 1st                            | Target touching, placement     | Delay: $\leq$ 50 (B), 75, 150, 250, 300, 350   | SoE, Performance       | Performance at 250 ms, SoE at 350 ms       |
| Bovet et al. [6] (2018)         | Yes                  | Yes (full body) | 1st                            | Sitting in chair, self-contact | Offset: 0 (B), 7.5                             | SoA, SoBO              | SoA, SoBO at 7.5 cm                        |
| Waltemate et al. [46] (2016)    | No (CAVE)            | Yes (full body) | Mirror                         | Trace upper body pose          | Delay: 45 (B), 75, 125, 210, 350               | SoA, SoBO, Performance | SoA, SoBO at 125 ms, performance at 75 ms  |
| Franck et al. [14] (2001)       | No (desktop monitor) | No              | Mirror                         | Move joystick with hand        | Delay: 1 (B), 50, 100, 150, 200, 300, 400, 500 | SoA                    | SoA at 150 ms                              |
| Groen and Werkhoven [18] (1998) | Yes                  | Hands only      | 1st                            | Object manipulation            | Offset: 0 (B), 10                              | Performance            | No impact                                  |

## 2.1 Self-avatars in VR

A short section on self-avatars in VR is included to familiarize the reader with this concept. In this paper when we mention virtual reality (VR), we refer to fully immersive VR where the user wears a head-mounted display (HMD) tracked in six degrees of freedom (6DoF). Which means both its position and rotation are tracked along three orthogonal axes. We make this distinction because, in older literature, VR may refer to any computer-generated environment [3], and is sometimes divided into categories of non-immersive: desktop screen, semi-immersive: cover more field-of-view than a typical desktop screen (e.g. a large concave screen), and fully immersive: blocks out the physical environment and replaces it with a virtual environment (e.g. a HMD).

A self-avatar in VR is a representation of the user in the virtual environment. When we write self-avatar in this paper, we refer to a humanoid avatar of at least the upper body (torso, arms, and hands). The head is irrelevant unless the avatar is shown in a mirror to the user. However, most studies on self-avatars discussed in this paper use full-body avatars.

To have a positive effect on the user’s feeling of being present in the virtual environment, it is important that it is centered at the user’s position, and that the motion of the self-avatar is as accurate as possible to the user’s real motion [15, 11]. It is also desirable to calibrate the self-avatar such that its limb lengths match the user. Centering the avatar at the user’s position is not a problem since the position and orientation of the HMD are available. Tracking the rest of the body can be done with motion capture setups. The two main types of motion capture use either IMUs (Inertial Measurement Units) or optical markers. Optical motion capture often produces more precise tracking but can have issues with occlusion, while IMU motion capture is known to have problems with drift [19]. Motion tracking setups often use dozens of sensors or markers and can be quite expensive. Consumer VR hardware usually only includes an HMD and a 6DOF motion controller for each hand. However, additional hardware specific to VR can be bought for tracking additional points on the body, e.g. HTC Vive trackers<sup>1</sup>. In the case of such sparse tracking, where several points on the body are not tracked, one usually tries to estimate the position and orientation of those missing points. Traditionally, this has been done using inverse kinematics [2]. Caserman et al. [8] recently did a review on motion reconstruction methods for self-avatars in VR. Newer approaches are being worked on that use data-driven methods [12].

In terms of how motion capture compares to solutions that use estimations, Debarba et al. [11] found that a virtual self-avatar with eight trackers and upper and lower body animation via inverse kinematics could perform as well as a 53-marker tracking setup in terms of feeling in control of the avatar. Disregarding HMD and motion controller tracking, participants found the tracking of the feet, followed by the mouth and the fingers to add the most sense of control. A similar study by Eubanks et al. [13] compared tracking setups in use with inverse kinematics (no avatar, head+hands, head+hands+feet, head+hands+feet+pelvis)

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<sup>1</sup>Vive tracker [vive.com/us/accessory/tracker3/](https://vive.com/us/accessory/tracker3/) last accessed: 2024-05-16.

and found that tracking the head, hands, and feet significantly increased both embodiment and presence. These results might hold in general, but what motion the user is expected to perform with the self-avatar is also important to consider when deciding what positions on the body to track.

Except contributing to the feeling of presence in virtual environments, having a self-avatar has been found for instance to aid in distance judgment in virtual environments [30], and reduce cognitive load in memorization tasks [41, 33]. In the consumer space, self-avatars are popular in a variety of VR applications. Bonelab<sup>2</sup> and Blade and Sorcery<sup>3</sup> are two action games that place large emphasis on their self-avatars. In social VR applications such as VRChat<sup>4</sup> where a large number of people share the same virtual space, the self-avatar is usually a central part of the whole experience. Users put considerable amounts of time into customizing them, and many advanced users invest in sophisticated full-body tracking.

## 2.2 Sense of agency over self-avatars

In this section we present the sense of agency (SoA), both as a subject on its own, but also in relation to self-avatars in VR. The sense of agency is widely studied by itself. A recent review can be found by Wen and Imamizu [49].

### 2.2.1 Sense of agency

As written by Wen and Imamizu "The sense of agency refers to the subjective feeling of controlling one's own actions, and through them, external events.". It is the sense that one is the agent of their actions and the events caused by them. In contrast, SoA does not occur when it is evident that someone else is the cause of an action or event. In patients with schizophrenia, loss of agency over their behavior can happen, and they can have issues differentiating between consequences caused by themselves and consequences caused by others [38, 17]. Furthermore, there is evidence that the attribution of agency over actions and events caused by oneself compared to externally caused events have different neural correlates [40].

Wen and Imamizu describes three underlying aspects of the SoA: intention, action, and effect. Where the intention is referred to as the desired state after acting, action is referred to as being aware of one's action, and effect is referred to as the consequences of one's action. The functions involved in the body's sensorimotor system are directly linked to the action-effect process. For example, when reaching for an object, the visual system (part of the sensorimotor system) identifies its location, the brain plans the movement (involving both sensory and motor areas), and the motor system executes the reach. As the move proceeds, proprioceptive feedback informs the brain about the position

<sup>2</sup>Bonelab [store.steampowered.com/app/1592190/BONELAB/](https://store.steampowered.com/app/1592190/BONELAB/) last accessed: 2024-05-16.

<sup>3</sup>Blade and Sorcery [store.steampowered.com/app/629730/Blade\\_and\\_Sorcery/](https://store.steampowered.com/app/629730/Blade_and_Sorcery/) last accessed: 2024-05-16.

<sup>4</sup>VRChat [hello.vrchat.com/](https://hello.vrchat.com/) last accessed: 2024-05-16.

of the arm, allowing for adjustments to ensure accuracy. Any discrepancies between the feedback provided by the different systems involved in the sensorimotor system can reduce the SoA over an action. For instance, if the visual feedback tells the brain that your arm is in one place, and the proprioceptive feedback tells it that it is in another place, there is a mismatch in visuomotor feedback, which can negatively affect the SoA [46]. A sense of agency can also be felt towards external objects, such as when a player controls a video game character.

To clarify some terminology from the previous paragraph. Proprioceptive feedback is the internal feedback generated by neurons in joints, tendons, and muscles about the body and its limbs’ relative position and velocity in space [45]. It plays an integral part in providing motor control. In this paper, when we write *sensorimotor* feedback, we refer to the integrated feedback from the sensory systems (visual, auditory, somatosensory, and vestibular), and motor control. When we write that there is a *mismatch* in for instance visuomotor feedback, we refer to when the feedback from the visual system does not match the feedback from the system for motor control (e.g. proprioceptive feedback).

However, the SoA is not only affected by discrepancies in sensorimotor feedback but also cognitive factors such as perceived task performance [1], as well as mental and physical effort [23]. For instance, Aoyagi et al. [1] conducted a study where participants traced a path using a virtual cursor, a sphere (no self-avatar). The cursor was mapped to the participant’s hand position. They measured the task performance as deviations from the path. In one of the conditions, they applied an offset that corrected the virtual cursor’s position towards the path without the participant knowing. Surprisingly, participants reported a significantly higher SoA in this condition than in the condition where the cursor was more accurate to their real hand position. Findings such as this one suggest that the SoA has different levels: sensorimotor, perceptual, and cognitive levels [31, 42]. The most widely accepted model of the SoA is the comparator model. We describe it in the next Section 2.2.2.

The SoA has been shown to function as an internal reward for voluntary motor control, influencing our behavior and actions [25]. When we experience a strong SoA, it reinforces the selection of actions by enhancing our confidence in our ability to control outcomes. This reinforcement mechanism encourages us to continue selecting similar actions in the future, giving rise to a positive feedback loop. The same mechanism promotes fluency in those actions. Thus, SoA is important to promote motor learning for instance in patients with motor disabilities.

### 2.2.2 The comparator model of agency

We describe the comparator model within the scope of this paper. For a more detailed account of the mechanisms involved in the comparator model, we refer the reader to the following papers [51, 29, 4, 16].

The comparator model of agency explains the SoA through a mechanism of internal comparisons between predicted and actual sensory feedback of actions

[49]. According to this model, when we intend to perform an action (via a motor command), our brain generates a prediction of the expected sensory feedback via something called a forward model. This prediction is then compared to the actual sensory feedback received after the action is executed. If the predicted and actual feedback match, a SoA is experienced, reinforcing the feeling that we are the cause of our actions. Conversely, a mismatch between predicted and actual feedback can lead to a diminished SoA, as it suggests that the action or its effects were not entirely under our control. Prediction errors are fed back to the internal model for motor control such that it can be updated and reduce errors in the future. The comparator model highlights the importance of sensorimotor processes in forming the SoA. However, it does not give a clear account of how to link cognitive expectations (such as expected performance in a task) to the SoA. Figure 2.1 displays an overview of the comparator model.

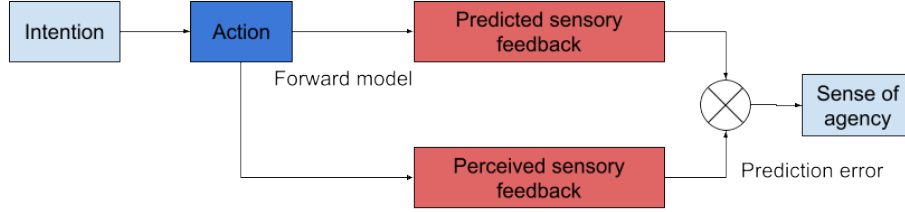


Figure 2.1. Displays the comparator model of agency. The comparator model of agency explains the SoA through a mechanism of internal comparisons between predicted and actual sensory feedback of actions. This figure is an adaptation of the one displayed in [49] (Fig. 5a).

### 2.2.3 Relation to self-avatars

The SoA is directly linked to controlling one’s own body, since doing so heavily reinforces a subjective feeling of control over one’s actions. This gives a natural connection to self-avatars in VR since they act as a representation of one’s own body in the virtual environment. Having a high SoA over the self-avatar is therefore important to promote action and interaction within the virtual environment.

Feeling like we have control over our body is largely dictated by whether or not we have actual motor control. Kilteni et al. [27] even defines the SoA as the sensation of having full motor control of one’s body, both causing and controlling its movements. Thus, to induce SoA over the self-avatar it is critical that the user’s body motion is reflected in the self-avatar motion, reinforcing the feeling that they have motor control over it. Therefore, when the user moves, the visual feedback of the self-avatar’s motion provided by the HMD must match the user’s internal proprioceptive feedback, ensuring visuomotor congruence. It could be argued that the vestibular system also is involved since it creates a sense of spatial orientation. Additionally, if an application with a self-avatar utilizes spatial audio, or haptic feedback, in combination with its visuals. Then, to achieve high SoA over the self-avatar it should try to ensure

full sensorimotor congruence both when controlling the self-avatar in isolation, but also when used to interact with the virtual environment. Discrepancies between the expected feedback when a user moves or interacts with something, and the actual feedback from any of these sensory modalities, can disrupt the SoA (as explained by the comparator model). However, to what extent the SoA is impacted is not clear and may depend on several factors including cognitive ones. Cognitive factors such as perceived performance in various tasks have been shown to have a significant influence on the SoA in studies both with self-avatars [46] and without [1]. In Section 2.3 we examine previous work on how motion delay and position offsets in the self-avatar can affect the SoA.

In the context of self-avatars in VR, the SoA is closely related to another concept: the sense of embodiment (SoE). In [27], the SoE refers specifically to embodiment of virtual self-avatars in VR and is made up of three components: SoA, sense of body ownership (SoBO), and sense of self-location. Note that these three are distinct concepts. We address the SoE since much work on virtual embodiment using self-avatars refers to Kiltner et al.. Sense of body ownership stems from the well-known experiment of the rubber hand illusion described in the 1998 paper by Botvinick and Cohen [5]. The body ownership illusion is the phenomenon where individuals perceive a fake body part or an external object as part of their own body. Several studies have investigated the body ownership illusion using full bodies in VR and have shown that it is possible to induce full-body ownership illusion to some degree [32]. Regarding the relation between SoA and SoBO - they can be experienced independently of one another, but SoA seems to be higher when there is also body ownership illusion [24]. Not unlike the SoA, congruent sensorimotor feedback is important to maintain SoBO as well [44, 37, 26]. The sense of self-location refers to the feeling that the virtual body is located in the same space as your physical body. It is usually not included in studies on self-avatars. In terms of the relative importance of SoA, SoBO, and sense of self-location, on the SoE, Fribourg et al. [15] found that people tend to prefer adjusting the point of view to coincide with the avatar, as well as increasing the quality of control, before adjusting the avatar’s appearance.

Studies about self-avatars in VR measure the SoA to indicate how well their self-avatar motion performs concerning the user’s subjective experience [7, 52]. Additionally, the SoA is measured to examine how deliberate mismatches in sensorimotor feedback, such as motion artifacts in the self-avatar, affect the user’s experience [46, 6, 28, 36, 1]. However, the SoA over self-avatars is often measured solely as a way to measure subjective control, but as we’ve discussed, it also gives important indications for voluntary action within the virtual environment. Furthermore, studies that measure the SoA often also measure the SoBO, but rarely the sense of self-location. Kalckert and Ehrsson [24] showed that SoA and SoBO correspond to different cognitive processes. To reduce the effect of individual criteria, the SoA can be measured implicitly via its effects on perception and behavior [49]. However, that is outside the scope of this study. Studies on self-avatars regularly use explicit measurements (usually questionnaires) to measure the SoA. Kiltner et al. suggests several statements for measuring the

components in the SoE (which have seen repeated use in previous literature). Roth and Latoschik [36] constructed a new qualitative measurement device for virtual embodiment, called the Virtual Embodiment Questionnaire (VEQ), used for instance in [52]. Example statements for SoA are "I felt like I was causing the movements of the virtual body", and "I felt like I was controlling the movements of the virtual body". Example statements for SoBO are "It felt like the virtual body was my body." and "It felt like the virtual body belonged to me". The present study uses the agency component of the VEQ, and two statements from the body ownership component, to measure SoA and SoBO (see questionnaire Q in Table 3.3). Good task performance in tasks that involve complicated motion can be seen as an indication of high SoA under the assumption that motor tasks are performed more successfully if the self-avatar executing the task is under finer control [27]. Therefore, task performance is sometimes measured as a complement to SoA in self-avatar studies.

## 2.3 Related work

In this section, we present relevant related work on the impact of delay, offset, and some other motion artifacts, on mainly the sense of agency (SoA) and task performance in various tasks, mostly in the context of self-avatars in VR. If the study did not use VR or a self-avatar it is explicitly noted. For an overview, see Table 2.1.

### 2.3.1 Motion delay

Delay is present as latency in all computing systems where data is processed and transferred. Some motion tracking systems such as those built into VR hardware for the controllers and the HMD, usually have lower latency. While external systems used for motion capture tend to have higher latency both due to processing demands and sending data over a network. In virtual self-avatars, delay between user motion and the corresponding visual feedback in the avatar is known to reduce the SoA. Adding delay between action and effect is often used to disrupt the SoA [48]. However, at which magnitude it affects the SoA depends on factors such as the task, whether or not the user has to focus on their motion, and the point of view from which the user sees their self-avatar [46, 28].

In a non-VR study where participants manipulated a joystick and saw a corresponding video of a hand mimicking the movements at certain delay values, Franck et al. [14] found that delay affected the perceived sense that the participant was causing the action at 150 ms (delay conditions in ms: 50, 100, 150, 200, 300, 400, 500).

Waltemate et al. [46] found that in a task where users see their self-avatar in a mirror, using a CAVE, and have to follow a moving pose of another avatar with their upper body while staying within certain bounds of the other avatar's limbs, the perceived SoA was negatively affected at 125 ms delay (delay conditions in ms: baseline=45, 75, 125, 210, 350) while user performance was

negatively affected at 75 ms. Measured SoBO was also affected first at 125 ms. All were further worsened at higher delays. A CAVE (Cave Automatic Virtual Environment) is a room with projections on the walls that display the virtual environment. In the context of self-avatars, an important difference between a CAVE and an HMD is that a CAVE does not hide the user’s real body, and neither does it block out the physical environment fully. They found in their analysis that user task performance predicted responses to both SoA and SoBO statements (both SoA and SoBO used one binary scale statement each). However, they note that the visual feedback provided to users on their task performance was quite strong, which can have affected this. Regardless, this suggests that smaller delays can be perceived if they cause performance errors (that are well-mediated to the user). Therefore, the difficulty of the task is important to consider. The same delay in the same VR system might be non-perceived in an easier task but perceived in a harder task. As we’ve seen, Aoyagi et al. [1] provided further evidence that it is the *perceived* task performance that adds to the SoA. Waltemate et al. also found that SoA and SoBO did not completely collapse at 350 ms, which they reason is likely because a full-body controlled self-avatar induces SoA and SoBO even with delays. On a separate note, their results showed that the reported SoA and SoBO were positively correlated with certain immersive tendencies among the participants.

Toothman and Neff [43] investigated the effect of several motion artifacts on social presence and the SoE. They found that in a task where users had to touch specific points or place objects accurately with their virtual hands while viewing their self-avatar from a first-person point of view, motion delay harmed the SoE first at 350 ms. The SoA was not analyzed individually, and the SoE metric was only three statements and a smaller focus of the study. Task performance (time and placement error) was affected first at 250 ms. Compared to the task in Waltemate et al., these tasks do not seem as sensitive to delays. Presumably, because they only focused on hand movement, which means the user only needs to adapt to the delays in a single body part to perform the task well.

Koiliias et al. [28] investigated the effect of several motion artifacts on the SoA from three different observation modes: first-person point-of-view, seeing the self-avatar in a mirror, and first-person while walking. The participant’s task was simply to observe the avatar’s motion while moving arbitrarily. None of the motion artifacts impacted the SoBO. Their delay condition was 150 ms and they found that it had a negative effect on the SoA in the first person point-of-view, and in the mirror observation mode, but not in first person point-of-view while walking. The authors attribute this to that the participants focus their gaze on where they are going next, a specific point in the room, and pay little attention to their virtual body’s motion. This might be partly because of the locomotion task, but arguably also because their specific task does not involve motions that require any form of awareness of where their limbs are positioned in space. They do address this, and note that participants said that since there were no obstacles in the path, they did not need to consider potential collisions. Therefore, not needing to give thought to their body composition while walking. Furthermore, they found that the SoA was most sensitive to motion artifacts in

the mirror observation mode. An explanation given by the authors is that the participants could view their self-avatar from different angles in the mirror, and gain a holistic view of the body’s composition, and therefore more easily single out details in the avatar’s pose that made it look wrong.

Roth and Latoschik [36] found delay to affect the SoA and SoBO from between 91 ms and 353 ms. Delay conditions in ms (baseline=90, 206, 353). Their study used a mix of first-person point-of-view and a mirror. The task was to follow movements as explained in voice instructions. It is not clear why the SoA was first affected by such a high delay in their study. Possibly, it had to do with the task.

Yun et al. [52] investigated how animation quality in terms of tracking solution and pose accuracy affected the SoA, and motor task performance. When comparing an IMU motion capture solution (91 ms) to two inverse kinematics-based solutions (32 ms and 33 ms), they found that the better inverse kinematics solution performed as well as the motion capture solution in tasks where the participants had to avoid obstacles with their arms or feet. However, in a task where the participants had to accurately match a static pose, the motion capture was significantly better. Across all tasks, the reported SoA was the same in the best inverse kinematics solution as in the motion capture solution. The authors suspect this was because of the delay in the IMU solution, but also because it had issues with sensor drift. This suggests that even delays smaller than 125 ms (as found in [46]) could affect the SoA. Since the pose accuracy was best using motion capture, but SoA could still be high using sophisticated inverse kinematics that performed well in the tasks. Yun et al. reason the SoA is not only related to the pose and motion of self-avatar but also to how it interacts with the environment (e.g. in their tasks). Based on cognitive factors affecting the SoA (Section 2.2.1), this is reasonable. Apart from the findings, Yun et al. puts forth a solid methodology of how to study user performance combined with the SoA, using tasks where the user concentrates on their body motion in interactions with the environment. The authors recognize the need to investigate SoA in tasks where the user not only moves the self-avatar in isolation, but also uses it to interact with the virtual environment. Our study uses an adapted version of their methods to examine the effect motion delays and offsets on the SoA in an interaction task.

These findings on how motion delay affects the SoA suggest that the task and view of the self-avatar have considerable influence. In tasks where the user does not have to be aware of their body motion and focus on other things, delay does not necessarily have a negative impact. However, the more the user has to be aware of the motion of body parts involved in completing a task, the higher the impact on the SoA. Such tasks are those that require high motor control. This is reasonable since mismatches in sensorimotor feedback would then be more apparent. Furthermore, perceived task performance adds or subtracts from the SoA. Therefore, how one displays performance feedback can affect the SoA. The lowest delay threshold found so far which affects the SoA in self-avatars is 125 ms (compared to 45 ms baseline). However, if the baseline is decreased in combination with other types of tasks, for instance, those where more interaction

with the virtual environment is required, lower thresholds could be found. We examine this in the present study. Our motion delay conditions (see MRQ1 in Section 1.1) were informed by the listed prior findings.

### 2.3.2 Position offset

Offsets between the tracked position of an object and its actual position can occur for instance because of inaccurate self-avatar dimensions or calibration, poor calibration in motion tracking setups, or drift when using IMU trackers [19, 52]. Few recent studies have been made that directly examine the position offset of body parts with the SoA.

One study that investigated this was conducted by Groen and Werkhoven [18] in 1998. In their study, the participant’s hand position was altered using a 10 cm lateral offset while they were immersed in a virtual environment. No effect on manipulation performance was found when participants had to move blocks. However, a negative aftereffect was found: participants adjusted their hand position 20 % of the offset in the opposite direction after exiting the virtual environment. They reason that participants can quickly adapt to a static offset in their hand positions, which could be why manipulation performance was not affected significantly. Held et al. [21] (1966) found that such adaptation does not happen above motion delays of 300 ms (no VR, no self-avatar). In later years, Held and Durlach [20] (1993) found that motion delays above 60 ms disrupt the sensorimotor adaptation (no VR, no self-avatar). Their research revolved around tele-operating robots, where human operator adaptation to both motion delays and position offsets is important to be knowledgeable about.

Bovet et al. [6] examined how position offsets affect SoA and SoBO in a task that involves self-contact. They found that an offset of 7.5 cm significantly reduced both metrics. They also included another motion distortion and confirmed that as long as the self-avatar visual was congruent with the self-contact tactile feedback, the SoA and SoBO remained high. This suggests that aligned self-contact between the user’s real body and the self-avatar is critical to the overall SoE. In this sense, any position offset in self-avatar body parts can be disruptive.

Porssut et al. [35] recognized that other forms of internal feedback similar to self-contact could be an important contributor to the SoE. They examined the impact of reaching articular limits in joints. In particular when performing a reaching motion and fully extending the arms. They found that SoE was impacted negatively if the self-avatar arms didn’t coincide with the user’s arms when they were fully extended and suspected this was because fully extending your arms gives an additional internal proprioceptive feedback, which if not matched visually in the avatar, gives rise to a mismatch that reduces the SoE. Offsets in self-avatar body parts can cause the virtual arms to not extend fully when the user extends their arms, which would then reduce SoE.

As mentioned in Section 2.3.1, Koiliias et al. [28] investigated several motion artifacts on the SoA. No static offset was tested, but temporarily applied offsets (20 cm, 16 Hz) were (also known as ”popping”), and caused a decline in the

SoA. This type of popping was motivated by signal noise in IMUs. They note that it was exaggerated in their study and that depending on the severity, signal noise can be counteracted by smoothing algorithms.

The findings presented in these studies and the one by Aoyagi et al. [1] (Section 2.2.1) suggest that how offset affects the SoA and task performance depends on whether or not there is delay and on what motion the user performs, which depends on the task. After a certain threshold of motion delay, users are not able to adapt to the offset, which reduces task performance, and presumably SoA. When there is no delay, offset has not been shown to affect task performance in manipulation tasks. In tasks where self-contact is involved, small offsets can affect SoA. In tasks where the user reaches articular limits in their joints, offsets can affect the SoE, and presumably the SoA. However, if the offset aids the user in their task, it can instead increase the SoA. In our task, no self-contact is necessary, therefore it is interesting to investigate whether or not our offsets have an effect on the SoA. Our offset conditions (see MRQ2 in Section 1.1) were informed by the listed prior findings. We could not find any prior research on the effect of combinations of motion delay and offset on the SoA. But, based on the finding by Held and Durlach on task performance, SoA could be affected in a similar way. Our mixed conditions (see MRQ3 in Section 1.1) are simply combinations of particular delays and offsets chosen from our delay and offset conditions.

### 3 Methodology

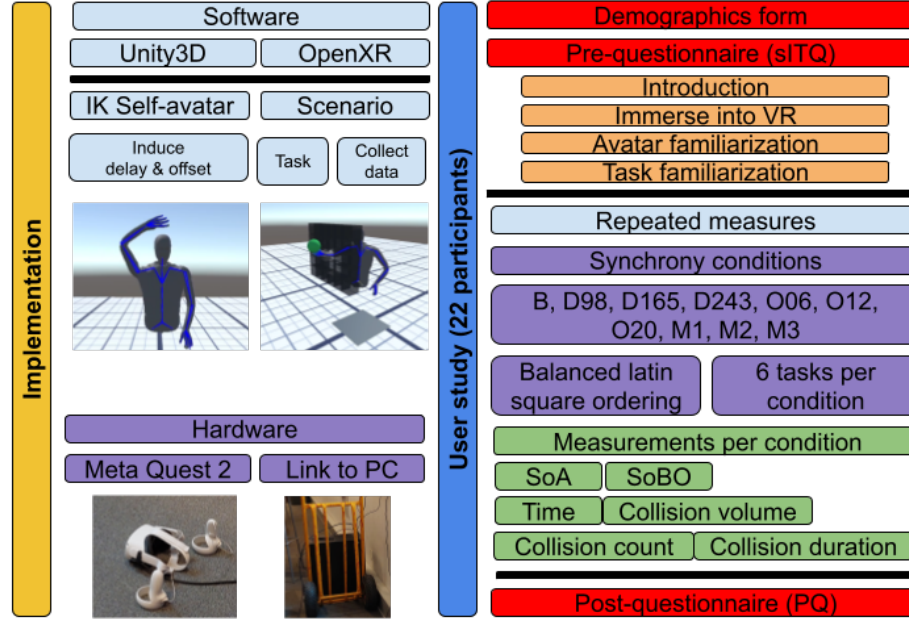


Figure 3.1. Displays an overview of the methodology.

An overview of the methodology is shown in Figure 3.1. The purpose of this study was to investigate the effect of delay and offset applied to the movements of a self-avatar on users' performance and sense of agency (SoA) when performing tasks that require precise interactions with the virtual environment. A virtual reality prototype application was developed that contains an upper-body self-avatar animated using inverse kinematics (IK), a scenario where users perform manipulation tasks, quantitative and qualitative measurements, as well as functionality to induce delay and offset on the self-avatar's motion. A repeated measures user study was designed where participants performed a set of manipulation tasks (also called pick-and-place tasks) for each synchrony condition.

#### 3.1 Implementation

The virtual reality application prototype was developed using Unity3D<sup>5</sup> (version 2021.3.25f1). Unity3D is a game engine and has extensive XR support through a range of XR plugin packages. We used the OpenXR Plugin package. OpenXR<sup>6</sup> is an open-source and royalty-free standard that aims to allow developers to build XR applications that work on a wide range of XR devices. In our

<sup>5</sup>Unity3D [unity.com](https://unity.com) last accessed: 2024-04-25.

<sup>6</sup>OpenXR [khronos.org/openxr/](https://khronos.org/openxr/) last accessed: 2023-11-20.

case, Valve SteamVR<sup>7</sup> was used to provide the OpenXR runtime. Our target device was a Meta Quest 2<sup>8</sup> running via Quest Link<sup>9</sup> cable to a desktop PC with the Meta Quest software (to enable Quest Link), as well as two Quest 2 motion controllers. However, since we used OpenXR, it should run on other XR hardware as well. The next subsection contains a brief introduction to essential concepts in Unity3D.

### 3.1.1 Setting up a scene for VR in Unity3D

A scene in Unity3D can be viewed as a container for all entities that relate to a specific part of a game. For instance, different scenes can be used for different levels. On a basic level, a scene contains the main camera, a light source, and maybe some geometry. Our prototype uses a single scene that contains the self-avatar, the user study scenario, a control interface, and all related logic. All entities in a scene are contained in a scene graph, a hierarchical data structure that defines the parent-child relationships between entities. All transformations to a parent entity, translation, rotation, and scale, affect its children as well. Each entity in a scene is a `GameObject`. `GameObjects` can have components attached to them, which define appearance and behavior. Component is the base class for everything attached to a `GameObject`. Example components are the Transform (describes position, rotation, and scale), and the MeshRenderer (renders the mesh). The Unity3D coordinate system is left-handed, with the positive x-axis pointing to the right, the positive y-axis pointing up, and the positive z-axis pointing forward. To create custom `GameObject` behaviors, for instance, to control user input, or respond to gameplay events, one can use C-sharp (programming language) scripting. Scripts are applied to `GameObjects` as components and can access other components and communicate with other scripts. Though, it is also possible to write C-sharp scripts that are not components.

There are several ways to set up a VR scene in Unity, depending on which XR plugin and packages one uses. As mentioned, our project used the OpenXR Plugin package. Additionally, we used the XR Interaction Toolkit<sup>10</sup> (XRI) (version 2.3.2) which is a package from Unity3D that provides a framework for working with interactions in XR. We used the XR Origin prefab that comes with the XR Core utilities (included as part of XRI) to handle all tracking and interfacing with the VR hardware. A prefab is a pre-configured `GameObject`. The XR Origin `GameObject` contains a main camera which is set to the VR headset's tracked position and rotation, and `GameObjects` for each VR controller that match their tracked position and rotation (see Figure 3.2). Our prototype uses the room-scale tracking mode, which means it uses a point on the floor of the

<sup>7</sup>SteamVR [steamvr.com](https://steamvr.com) last accessed: 2023-11-20.

<sup>8</sup>Meta Quest 2 [meta.com/se/en/quest/products/quest-2/](https://meta.com/se/en/quest/products/quest-2/) last accessed: 2023-11-20.

<sup>9</sup>Quest Link [meta.com/en-gb/help/quest/articles/headsets-and-accessories/oculus-link/connect-link-with-quest-2/](https://meta.com/en-gb/help/quest/articles/headsets-and-accessories/oculus-link/connect-link-with-quest-2/) last accessed: 2023-11-20.

<sup>10</sup>XR Interaction Toolkit [docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit@2.3/manual/index.html](https://docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit@2.3/manual/index.html) last accessed: 2023-11-21.

physical space as the tracking origin and supports 6 DoF tracking. At initialization, XR devices choose a point in the physical space as the origin, all tracking data is relative to this point. In our experiment there was a need to reposition the tracking origin to align the virtual environment with the physical space (a procedure called “recentering”), this functionality is provided by SteamVR.



Figure 3.2. Displays the XR Origin GameObject and its children in the scene graph.

To handle user input from the VR controllers we used the action-based input system provided by the XRI package. Actions are for instance “Select”, “Activate”, “UI Press”. Each action is mapped to an input binding. An input binding may be labeled e.g. “gripPressed”, and map to different buttons on different XR hardware. This enables easier targeting of multiple XR devices. The actions fire events that can be listened to and handled as seen fit. For our prototype, we used the default input action mappings provided by the XRI package. Exactly what these bindings are will be specified in their relevant context instead of here.

### 3.1.2 Self-avatar

The self-avatar consists of a 3D humanoid model, a definition for the bone hierarchy, an inverse kinematics solution to position and orient the arms, a simple algorithm for orienting the body’s forward direction, and a solution for resizing the arm lengths of the avatar to fit the user.

We used a generic 3D humanoid model. The model consists of a mesh and a bone hierarchy (also known as a rig). It was imported as a Unity3D GameObject that in addition contained a skinned mesh renderer component, and a humanoid avatar definition. Figure 3.3 shows the humanoid model, note that the legs are hidden since we are only concerned with the upper body. Unity3D uses a specific bone hierarchy for humanoid avatars. The bone hierarchy for the upper body is shown in Figure 3.4. It is used to create the humanoid avatar definition, which in turn is used as input to the animation system when animating the avatar.

To animate the avatar’s arms based on the tracked VR controllers and headset positions, we used the Animation Rigging package (version 1.1.1) which contains animation constraints among other things. The avatar GameObject center position is set to the VR headset position but with a downwards offset on the y-axis (up vector). To constrain the avatar’s head position to the VR headset position (resulting in a first-person view of the avatar), we used a Multi-Parent Constraint which simply lets the head bone of the avatar act as

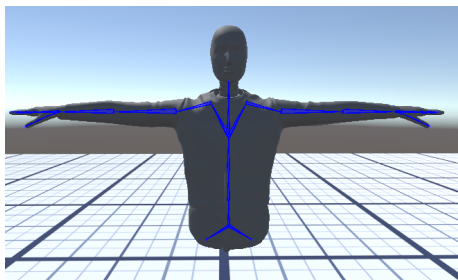


Figure 3.3. Displays the humanoid 3D model.

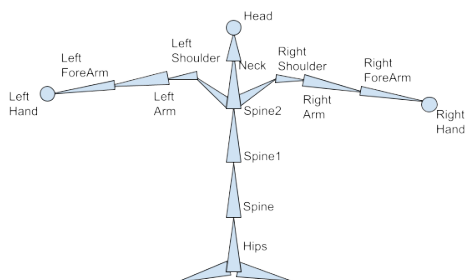


Figure 3.4. Displays the upper body bone hierarchy. Finger bones and leg bones are not shown.

a child of the VR headset GameObject without actually being a child in the scene graph. To animate the arms we use one Two Bone IK Constraint for each arm (Figure 3.5). This constraint takes a root, a mid, and a tip transform to describe the kinematic chain (three joints). The mid transform must be a child of the root, and the tip transform, which is the end effector (the last joint in a kinematic chain), must be a child of the mid transform. The constraint also takes a target transform and a hint position. It solves the IK in such a way that the mid transform is as close to the hint position as possible. The root transform for the right arm was set to “RightArm” (effectively the avatar’s shoulder position), the mid transform was set to “RightForeArm” (effectively the avatar’s elbow position), and the tip transform was set to “RightHand” (see Figure 3.4 for reference). The IK target was mapped to the position and rotation of the right VR hand controller, and the hint was positioned slightly to the right, backward, and down in relation to the avatar’s forward direction. The hint is used to control the elbow bend direction. The left arm IK was set up in the same way except it used the left arm transforms and controller position. Offsets were used to tune the IK target positions such that the avatar’s hands line up with the user’s hands while they hold the VR controllers (therefore, the IK targets are not positioned or oriented exactly as the VR controller, this is not shown in the figures since it would just make them convoluted).

We used a method inspired by Parger et al. [34] to determine the avatar’s forward direction based on the VR controllers’ position and the VR headset’s

position. We calculate the direction from the headset position to the middle of the hands' IK targets. The avatar's forward direction is then set to a weighted average of the headset's forward direction and that direction, as shown in Figure 3.6. The directions are only calculated in the xz-plane, which means the avatar only rotates around the y-axis. This is ok considering the user should not need to pitch or roll their upper body during the user study tasks. The avatar's forward direction is important since it determines the position of the avatar's shoulders which are the root transforms of the arm IK.

Since the user study has multiple people do the tasks there needs to be functionality to resize the avatar arm lengths to fit the user. It is not possible to resize the avatar by altering the bone positions or scale on the corresponding GameObjects in the scene graph. Altering these values during runtime does not affect the underlying animation system that calculates the animation constraints. To resize the avatar during runtime, and have the underlying animation system recognize it, we need to reset and provide a new humanoid avatar definition with the new and resized bone transform definitions via a script. This can be done during runtime by using the AvatarBuilder and HumanoidDescription classes. We resize the avatar's upper arms and forearms by repositioning the bone transforms of the right and left "ForeArm" and "Hand" transforms. The estimated arm lengths are calculated as the length from the avatar's shoulder position (left/right "Arm" bone) to the VR hand controller position. Figure 3.7 shows an example of the self-avatar matching the user's pose.

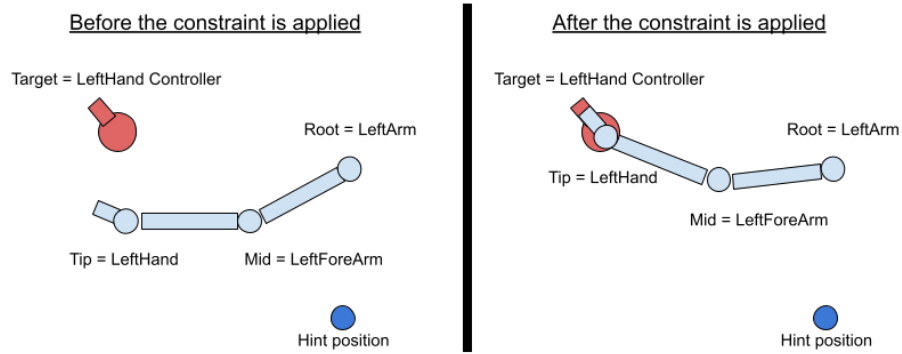


Figure 3.5. Displays the effect of the Two Bone IK Constraint in 2D. Each transform is represented by a disk (position) and a rectangle (orientation). The left image shows the arm pose (light blue) before the constraint is applied, and the right image shows the arm pose after the constraint is applied. The Two Bone IK Constraint calculates a feasible position and orientation for the Mid transform such that the Tip is positioned and oriented as the Target (red), and the Mid transform is positioned as close to the Hint position (dark blue) as possible, and the Root stays at its original position.

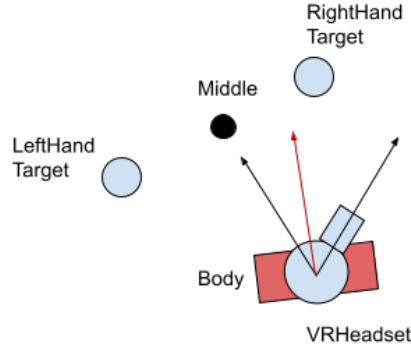


Figure 3.6. Displays the method for orienting the avatar's body forward direction. The body (red rectangle) forward direction (red vector) is a weighted average of the VRHeadset forward direction and the direction from the VRHeadset (light blue transform) to the Middle of the Target positions.

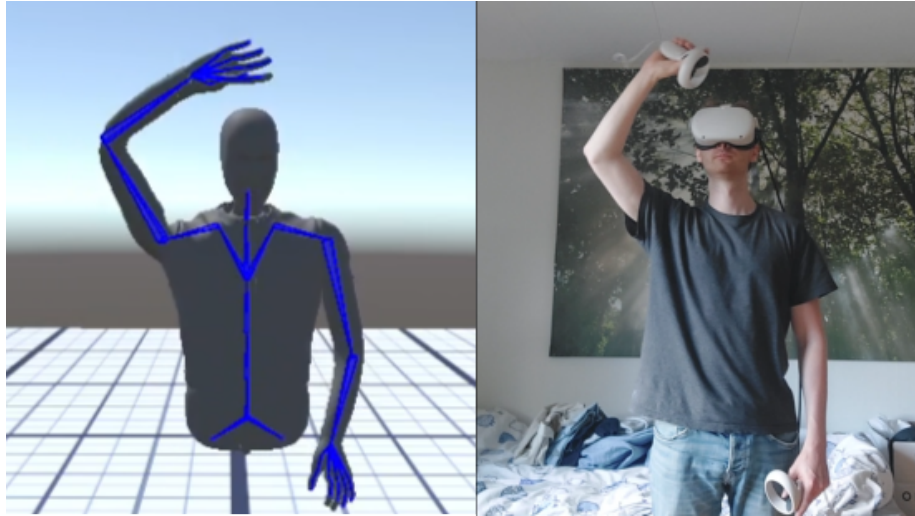


Figure 3.7. Displays the self-avatar matching the pose of the person wearing the VR headset and controllers. The head is angled slightly forward since that angle produced the least visual artifacts without removing the head entirely.

### 3.1.3 Inducing motion delay and position offset

The two independent variables in our experiment were delay applied to the motion of the self-avatar, and offset applied between the user's real hand positions and the self-avatar's hand positions. Both delay and offset are induced by controlling the transforms of the inverse kinematics targets for the Two Bone IK Constraint used to animate the avatar's arms. Figure 3.8 displays a visual of how both position offsets and motion delay were applied.

Delay in the avatar’s arm motion was induced in the following way. A desired delay is set. Let’s consider one arm. Each frame, the position and rotation of the VR hand controller are added to a queue data structure, along with a timestamp of when this tracking data was recorded. Each frame, we check the difference between the current time and the timestamp of the element at the front of the queue, and dequeue elements until the difference is less than the specified delay. We then apply the tracking data of the element that was last dequeued. By doing this, the queue that holds the tracking data to apply to the IK target is offset by the desired delay. This approach was inspired by Toothman and Neff [43]. Note that we can only poll and apply tracking data each frame. In Unity3D, XR applications must use vertical sync. The Quest 2 has a 90 Hz display refresh rate, thus the application runs in 90 frames per second (it manages to hit this target framerate without issues on our hardware). This means the temporal resolution of our actual induced delay values are multiples of  $1/90 = 0.0111\dots$  seconds, or 11.111... milliseconds (with some small variance in the decimals). E.g. if we set the desired delay to 50 ms (milliseconds), the actual delay will be approximately 55 ms. We’ve also measured the motion-to-photon delay (time between motion and the resulting image being rendered on the display) of our system to be on average 32 ms via the Oculus Debug Tool, which means our baseline delay is approximately 32 ms and all induced delay is on top of this. This motion-to-photon delay is in line with those reported in studies using similar hardware [47, 52]. In practice, setting a 50 ms delay in the application results in an 87 (32+55) ms effective delay between the user’s real-world arm motion and the arm motion of the self-avatar. Our conditions in Section 3.2.1 state the effective delays.

Inducing offsets between the user’s real hands and the avatar’s hands was accomplished by offsetting the IK targets for both hands from the VR controllers’ tracked positions by a specified value in Unity3D units. One Unity3D unit is one meter in the prototype. For each hand, the offset is set along the x-axis of the self-avatar’s local coordinate system (transformed to world space). They are mirrored such that the left arm IK target is offset to the left, and the right arm IK target is offset to the right. Inducing an offset causes the virtual arms to move outwards as a result. The tracking accuracy of the Quest 2 motion controllers has been measured to be on the order of millimeters, less than 1 centimeter (cm) [22].

#### 3.1.4 User study scenario objects and controls

The virtual environment contains the self-avatar, a floor with a grid texture to support depth perception, and a gray rectangle on the floor to show the user where they are supposed to stand during the experiment. The user study scenario consists of the user picking up objects and moving them to a target location, carefully but quickly, while avoiding collisions between their virtual arms and obstacles on the way to the target. Each task in the scenario consists of one pickup object, one target object, and a set of moving obstacles. A control interface was implemented to more easily be able to manage the user

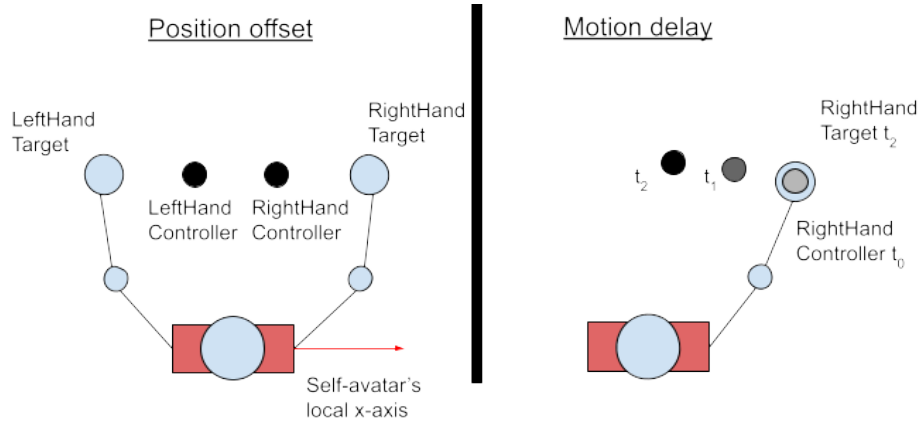


Figure 3.8. Displays a visual of position offset (left) and motion delay (right) when viewing the self-avatar from above. Position offset between the LeftHand Target and RightHand Target and the corresponding controller (black) was applied along the local x-axis (red vector) of the self-avatar. Negative values were used for the left target and positive values for the right target. Motion delay between the targets and controllers was applied by buffering the updates of the targets (only showing RightHand Target and LeftHand Controller). The black sphere shows the current position of the RightHand Controller, time  $t_2$ . At time  $t_2$  the RightHand Target is positioned where the RightHand Controller was positioned at  $t_0$ , showing it is two discrete time steps behind the controller.

study. Figure 3.9 shows the environment from a first-person view while a task is in progress.

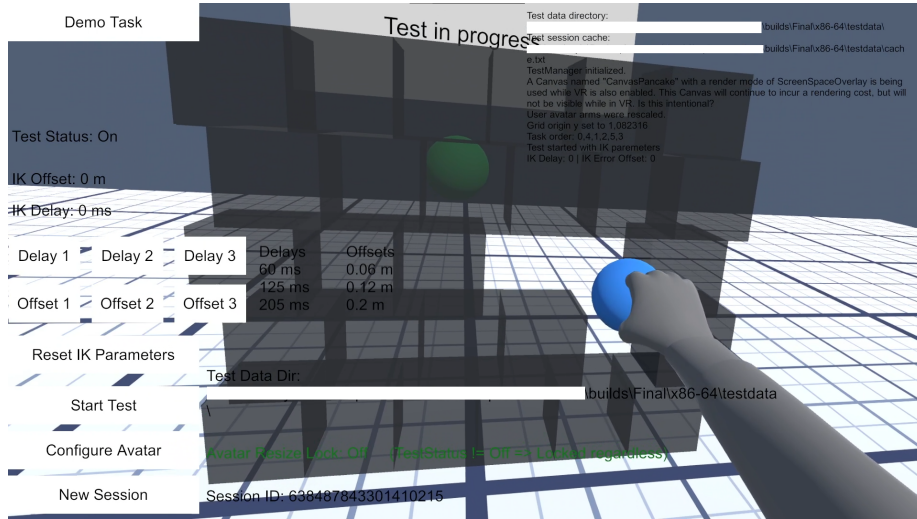


Figure 3.9. Displays the virtual environment while a task is in progress. The blue sphere is the pickup object, the green sphere is the target position, and the transparent blocks are the obstacles. The graphical desktop interface for controlling the scenario is also visible.

The user study tests (sequence of tasks followed by a questionnaire) are controlled by a manager script that can initialize tasks, set up the task order, collect measurements for each task, control the delay and offsets applied to the avatar, and keep track of the test status (e.g. if a task is in progress; if the questionnaire is in progress, etc). The controller GameObjects that are children of the XR Origin were set up with script components from the XRI package that enable interaction with world-space user interfaces, as well as interaction with GameObjects that have a corresponding script component that lets them be interacted with.

The pickup is a simple sphere GameObject with such a script component that lets the user grab the sphere using the “Select” action, which is mapped to the grip input button on the VR controllers. The component that controls interactions with the pickup was set up such that interaction is only possible when the hands of the avatar are in close vicinity to the pickup (regardless of VR controller position, which might not be the same when e.g. offset is applied). The target or goal is also a simple sphere GameObject which checks for collisions with the pickup. The obstacles are cube GameObjects with a multitude of smaller colliders inside of them to enable collision volume measurements (explained in Section 3.1.5). The obstacles move left to right, with obstacles on odd rows moving in the opposite direction of obstacles on even rows. The displacement is determined by a sine wave that takes the time since the task was started. All objects in a task are positioned on a 6x5x5 uniform grid with cube cells (15 cm in each dimension). A questionnaire interface (see Figure 3.10) in world space is implemented which the user can interact with while in VR using the “UI Press” action, mapped to the trigger input button on the VR controller.

The graphical control interface visible in Figure 3.9 lets the person controlling the experiment do a couple of useful things. The delay and offset values can be set, a new test can be started, and they can start and stop avatar configuration. Starting lets the user resize the avatar arms using the X input button on the left VR controller and stopping it locks the avatar resizing functionality. They can set a new session ID - the session ID is connected to the data collected during the current session and makes grouping of experiment data easier. There is also a button for starting a demo task to teach the participants how to perform the tasks. Additional functionality exists through key bindings, such as force-stopping a test. Figure 3.11 shows VR controller input and their use case in the application.

### 3.1.5 Collecting measurements

Four performance measurements are made for each task. The time it takes for the user to complete the task, the number of collisions between the user’s virtual arms and obstacles, the collision volume, and the total duration that the user is colliding. Qualitative measurements are made via the questionnaire described in the previous section.

When a task is spawned, only the pickup and the target are visible. The task and the measurements, start as soon as the user grabs the pickup. At this point,

Figure 3.10. Displays the questionnaire interface shown to the user in the virtual environment.



Figure 3.11. Displays VR controller inputs and their use case.

the obstacles become visible as well. Each arm has three capsule colliders, one for the upper arm, one for the forearm, and one for the hand as shown in Figure 3.12. Each obstacle is 15 cm x 15 cm x 15 cm in size ( $3375 \text{ cm}^3$ ), and consists of 125 sub-obstacles (children GameObjects) that are each 3 cm x 3 cm x 3 cm ( $27 \text{ cm}^3$ ). The obstacle has one box collider that covers the whole obstacle, and each sub-obstacle has its own box collider (see Figure 3.13). A cube with scale (1, 1, 1) in Unity3D is  $1 \text{ m}^3$ . One Unity3D unit is 1 m. Note that obstacles and cells in the grid have the same dimensions, leaving no gaps between placed obstacles.

The *task time* is the duration in seconds from when the user grabs the pickup and when the pickup collides with the target. An event is sent to the user study manager script when the pickup is grabbed, which sets a flag that a task is now running, and another event is sent when the pickup collides with the goal.

The *task collision count* is the number of collisions between the virtual arms and the obstacles. All three arm colliders for each arm act as the same collider in this case, and a collision is counted when any arm collides with any obstacle. For instance, if both arms collide at the same time with the same obstacle that is counted as two collisions, if one arm collides with two separate obstacles at the same time that counts as two collisions, and if one arm collides with one obstacle that counts as one collision. The virtual arms can pass through the obstacles,

the obstacles simply turn red if an arm is colliding. All collision measurements are reported by the obstacles to the user study manager script.

The *task collision volume* is measured by the sub-obstacles on each obstacle. Collision volume is increased by  $27 \text{ cm}^3$  each time an arm collider (any of the arms' upper arm collider, forearm collider, or hand collider) collides with any sub-obstacle collider. This way, a higher collision volume is measured when the user drags their full arm through an obstacle than when they only drag their hand through an obstacle.

The *task collision duration* is added to for each separate collision. A collision is determined in the same way as for task collision count. When a collision happens between an obstacle-arm pair, a timer is started for that pair. When all arm colliders for the colliding arm have left the obstacle, the duration is measured and reported to the user study manager script. The duration is also measured and reported if the task ends while an arm is currently colliding. For instance, if both arms collide with two separate obstacles each for five seconds, then the collision duration for that task is ten seconds.

When a test is started, the metadata for that test is written to a new file specific to that test (date and time, session id, delay and offsets, the shuffled task order). The task performance measurements for each completed task are written to the file, and so is the questionnaire results.

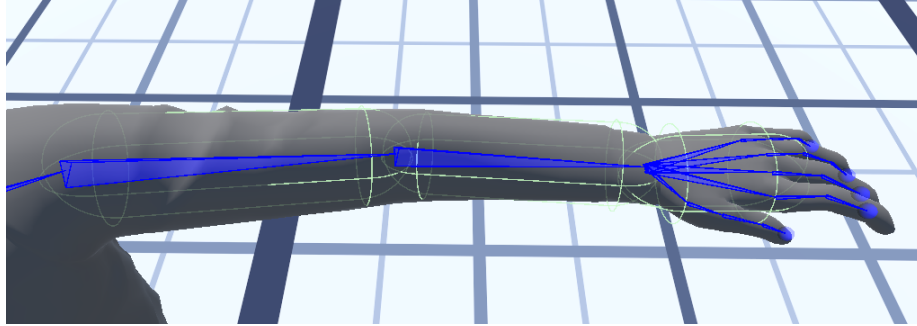


Figure 3.12. Displays the three capsule colliders (visualized as light-green outlines of the capsules) on the avatar's left arm, one for the upper arm, one for the forearm, and one for the hand (the collider for the hand is sized and positioned such that it aligns properly when the hand is closed and holding the pickup).

### 3.2 User study

The experiment used a repeated measures design to investigate how several delay and offset conditions on the self-avatar affect users' sense of agency (SoA), sense of body ownership (SoBO), and task performance in a pick-and-place task (sometimes called manipulation task in the paper) that required precise movement of the virtual arms. Participants also answered a pre-questionnaire on immersive tendencies, and a post-questionnaire related to the SoA. The tasks and the measurements were inspired by those used for the upper-body part

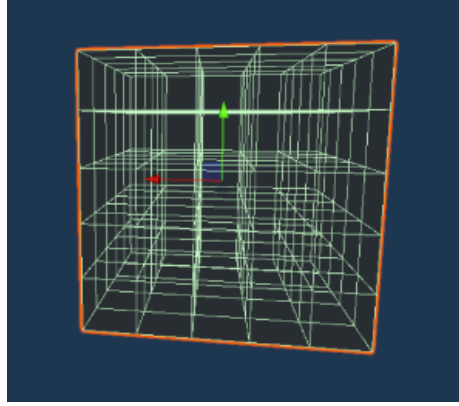


Figure 3.13. Displays an obstacle and all of its sub-obstacle colliders (box colliders are visualized as light-green outlines of the boxes).

of the study by Yun et al. [52]. The experiment followed a protocol where participants were introduced to the VR hardware and taught how to perform the tasks as intended. Each participant performed a set of pre-configured but randomly shuffled pick-and-place tasks for each experimental condition. For each condition, the tasks were followed by a questionnaire that measured both SoA and SoBO. The participants' averages per measurement for each condition were then analyzed via graphs and tests for significance (Section 4.1) between each of the delay conditions, each of the offset conditions, and each of the mixed conditions which combine both delay and offset. Two pilot studies were conducted to refine the protocol, tasks, and questionnaire statements.

### 3.2.1 Experimental conditions

There were ten experimental conditions in total (also referred to as synchrony conditions). Each condition varied the synchrony of the self-avatar's arms by inducing either delay, offset, or both. Delay caused the avatar's arm motion to be delayed by a set latency, and offset caused the avatar's hands (and thereby arms) to be offset by a set value. The baseline delay was 32 ms (milliseconds) and the baseline offset was  $< 1$  cm (centimeter). To not clutter the tables or the text, we specify the offset as 0 instead. The conditions were the following, where each tuple in the set represents a condition on the form (delay, offset): (32 ms, 0 m), (98 ms, 0 m), (165 ms, 0 m), (243 ms), (0 ms, 6 cm), (0 ms, 12 cm), (0 ms, 20 cm), (98 ms, 12 cm), (165 ms, 6 cm), (243 ms, 20 cm). See table 1 for the independent variables and conditions in tabular format. We name the conditions C1-C10, but also call (0 ms, 0 m) the baseline (B) condition, the delay conditions D98, D165, and D243, the offset conditions O06, O12, and O20, and the mixed conditions M1, M2, and M3. Table 3.1 shows all conditions. Delay and offset values were chosen based on previous work presented in Section 2.3, and results from our two pilot studies.

This study used a repeated measures design, which means that all participants experience all conditions. To reduce order effects, participant bias such as fatigue, and carryover effects between conditions, we used a balanced latin square to order the conditions for each participant. The latin square used is displayed in Table 3.2.

Table 3.1. Displays the naming, delay (milliseconds - ms), and offset (centimeters - cm), for each of the experimental conditions C1-C10.

|               | C1 (B)  | C2 (D98) | C3 (D165) | C4 (D243) | C5 (O06) | C6 (O12) | C7 (O20) |
|---------------|---------|----------|-----------|-----------|----------|----------|----------|
| <b>Delay</b>  | 32 ms   | 98 ms    | 165 ms    | 243 ms    | 32 ms    | 32 ms    | 32 ms    |
| <b>Offset</b> | 0 cm    | 0 m      | 0 m       | 0 m       | 6 cm     | 12 cm    | 20 cm    |
|               | C8 (M1) | C9 (M2)  | C10 (M3)  |           |          |          |          |
| <b>Delay</b>  | 98 ms   | 165 ms   | 243 ms    |           |          |          |          |
| <b>Offset</b> | 12 cm   | 6 m      | 20 m      |           |          |          |          |

Table 3.2. Displays the latin square used for ordering the conditions for each participant. Each row displays a condition order. The first column shows the expressions that determines which condition order the current participant number ( $i$ ) should use. Participant numbers start from zero.

| <b>Participant</b><br>( $i$ ) |     |     |     |     |     |     |     |     |     |     |
|-------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $i \bmod 10 \equiv 0$         | C1  | C2  | C10 | C3  | C9  | C4  | C8  | C5  | C7  | C6  |
| $i \bmod 10 \equiv 1$         | C2  | C3  | C1  | C4  | C10 | C5  | C9  | C6  | C8  | C7  |
| $i \bmod 10 \equiv 2$         | C3  | C4  | C2  | C5  | C1  | C6  | C10 | C7  | C9  | C8  |
| $i \bmod 10 \equiv 3$         | C4  | C5  | C3  | C6  | C2  | C7  | C1  | C8  | C10 | C9  |
| $i \bmod 10 \equiv 4$         | C5  | C6  | C4  | C7  | C3  | C8  | C2  | C9  | C1  | C10 |
| $i \bmod 10 \equiv 5$         | C6  | C7  | C5  | C8  | C4  | C9  | C3  | C10 | C2  | C1  |
| $i \bmod 10 \equiv 6$         | C7  | C8  | C6  | C9  | C5  | C10 | C4  | C1  | C3  | C2  |
| $i \bmod 10 \equiv 7$         | C8  | C9  | C7  | C10 | C6  | C1  | C5  | C2  | C4  | C3  |
| $i \bmod 10 \equiv 8$         | C9  | C10 | C8  | C1  | C7  | C2  | C6  | C3  | C5  | C4  |
| $i \bmod 10 \equiv 9$         | C10 | C1  | C9  | C2  | C8  | C3  | C7  | C4  | C6  | C5  |

### 3.2.2 Tasks

For each experimental condition (C1-C10), the participant performed six pick-and-place tasks (also referred to as just tasks), and then completed a questionnaire on the SoA and the SoBO. We refer to this as a *test*. The six pick-and-place tasks were randomly shuffled for every condition and participant to reduce ordering effects within each test.

A pick-and-place task consisted of a pickup object, a target object (also called the goal), and a set of moving obstacles that are arranged like a wall

with a gap. Six tasks were designed that differ in their positioning of obstacles, pickup, and target. Objects in each task were located on a 6x5x5 uniform grid with cube cells of size 15 cm in each dimension. The grid origin in world space was adjusted to fit the participant's height. Common to all tasks was that the obstacles were located on grid depth  $z=2$ , pickup on  $z=0$ , and target on  $z=4$ . The obstacles moved left and right on the grid x-axis with the displacement from their starting position determined by a sine wave that took the duration since the task was started. Obstacle movement was added (compared to the original task in [52]) to encourage participants' to make small continuous adjustments of their arm pose while performing the task, hopefully raising arm motion awareness. Obstacles on odd index rows moved in the opposite direction of obstacles on even index rows. Figure 3.14 displays a task. Task difficulty was tuned based on feedback from the two pilot studies. In all tasks, the wall gap and the goal were positioned such that the limitation of elbow bend direction (see Section 3.1.2) would not cause necessary collisions.

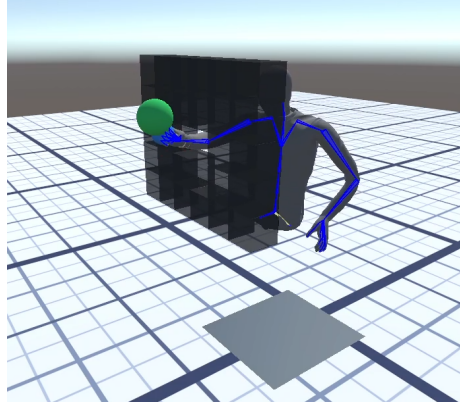


Figure 3.14. Displays a task being performed (Unity3D scene view). The blue sphere is the pickup object, the green sphere is the target object, and the transparent gray cubes are the obstacles. The self-avatar is holding the pickup and navigating it through the gap in the moving obstacles without colliding. Underneath the avatar is the gray floor plate (width: 65 cm, length: 75 cm).

Participants received the following *main instructions* for performing a task. “Your objective is to grab the blue sphere (the pickup), and bring it into the green sphere, the goal position, through the gap in the obstacle wall. Do this,

- as quickly as you can,
- while doing your best to avoid collisions between your virtual arms and the obstacles.
- If you collide, you should try to correct the collision by moving the arm out of the obstacle(s). I.e. you should try to keep your arms outside of the obstacles.”

They were also informed that the task does not start until they grab the pickup, that they can use any hand they like to grab the pickup, that they are only allowed to move their arms through the gap in the obstacles and not to the sides, over, under, or through the wall, and that they are allowed to move closer to the obstacles to more easily reach the target as long as they remain inside of the gray floor plate on the floor of the virtual environment, shown in Figure 3.14. Picking up and moving objects with their hands has been shown to make people pay more attention to their upper body [9], which is important here since we measure the SoA for that part of the body. Furthermore, we are interested in examining tasks where the user has to use their self-avatar to interact with the virtual environment, and have high awareness of their body motion. Based on previous work presented in Section 2.3, we believe this task will allow us to do that since it puts focus on high motor control.

In a test, the tasks were completed sequentially. The first task was spawned (only the pickup and target are visible) and the task started as soon as the participant used any of their virtual hands to grab the pickup, at which point the obstacles were made visible and the measurements started. The task was completed when the pickup reached the target (the participant heard an audible pop when they reached the target), and the second task was then immediately spawned. The second task started when the participant grabbed the new pickup. This continued until all six tasks were completed, after which the questionnaire was shown. Measurements are described in Section 3.2.5.

### 3.2.3 Apparatus

The user study was conducted in three separate areas in different rooms depending on room availability. The smallest area had the following dimensions: 210 cm x 240 cm, which were more than enough to let the participant move their arms freely. Figures display the experiment setup in the smallest area.

Two different Windows 10 64-bit machines were used, depending on the location. One machine had the following specifications: AMD Ryzen 7 3700X (3.6 GHz) CPU, RTX 3070 GPU (8 GB VRAM), and 16 GB of RAM. The other machine had the following specifications: AMD Ryzen 7 5800X (3.8 GHz) CPU, RTX 3080 Ti, and 32 GB of RAM. Both machines ran the application at a steady 90 fps.

The VR hardware used was a Meta Quest 2 (1920x1832 resolution per eye, field of view varies depending on the distance from the eyes to the lenses and interpupillary distance of the wearer) and two Meta Quest 2 controllers. As well as a USB3.0 to USB-C cable that connects the machine that runs the prototype to the VR headset- it is 6 meters long. The cable was used for Meta Quest Link, which enables running desktop VR applications on the Meta Quest 2 (which is standalone) through the Meta Quest Link software.



Figure 3.15. Displays the experiment setup in the room with the smallest area. Participants were positioned approximately where the VR headset is. The laptop was used for questionnaires. The bottom left corner shows the monitor with the experiment controls. The PC running the application was positioned on the yellow cart.

### 3.2.4 Procedure

The user study had 22 participants (5 female, 17 male, aged 18 to 78, age mean=30.09, age sd=14.86, VR experience mean=2.86, VR experience sd=1.14). A note on the age range and distribution: 18 of the participants were between ages 18 and 35, and three were between 45 and 55. As far as we are concerned, all participants, including the 78-year-old senior, understood the task and questionnaire statements. The following scale was used for the VR experience: 1 - I have never tried VR before, 2 - I have tried VR once, 3 - I have tried VR more than once, 4 - I use VR occasionally, 5 - I use VR frequently.

The demographics form used to collect this information was filled in by participants as soon as they arrived at the experiment location (one of the rooms used for the experiment is shown in Figure 3.15). Afterward, they filled out a pre-questionnaire on immersive tendencies (see Table 3.4). At the same time, we prepared the condition order for the current participant number (latin square in Table 3.2). Participants then got a short introduction to the study where it was explained that they were going to wear the Quest 2 VR headset and controllers, embody a self-avatar, and perform a manipulation task. The task and the main instructions were presented to the participants before they were immersed in VR such that they had a baseline understanding of it from the start. They were also informed about the number of tasks they would perform. We had them read the questionnaire (see Table 3.3) once before being immersed, since otherwise, they would have seen the statements before every condition except the first. This also gave them a short time to reflect on their meaning. The participants were also shown which button to grab the pickup with before being immersed.

Participants were then immersed in VR. The self-avatar was not visible at this point. After instructing how to properly equip the VR headset, we made sure that participants stood in a safe location such that they would not collide with any real-world objects while moving their arms. Then, we recentered the virtual environment, and informed the participants that as long as they stood on the gray floorplate in the virtual environment (Figure 3.14), they could move their arms freely without colliding with anything in the real environment. We then helped the participant calibrate the lengths of their virtual arms until they felt the correct length. Then, to familiarize the participants with the virtual arm motion, we asked them to move their virtual arms until we said stop. In the meantime, we enabled the best synchrony condition, as well as the worst delay and offset conditions in certain intervals, without telling the participant. Afterwards, we asked if they noticed anything concerning the arm motion, without confirming their observations. This was done to make the participants aware of the motion of the virtual arms, and to reduce the risk of them thinking there was an issue with the program. Finally, we double-checked that the lengths of the virtual arms were correct, and then we locked the resizing, which also disabled the visual of the avatar until a test or demo task was in progress.

Next, participants were taught how to perform the pick-and-place task. A special demo task was used for this which was not in the usual set of six tasks. The demo task always used the baseline condition. Participants performed demo tasks until we found that they had understood the instructions properly and performed the task as intended. Usually, participants performed between five and eight demo tasks. The first few demo tasks were used to teach the instructions in separation, while in the last two tasks, the participants were instructed to perform the demo task as if it were part of the real test. Participants usually had to be reminded that a task does not start until they grab the pickup, and that they are free to relax before grabbing it, but as soon as they grab it they should follow the main instructions (navigating quickly while trying to avoid collisions). This way of teaching the task substantially reduced initial strategy development seen in the first pilot, as well as removed a problem where participants in the pilot would test out the motion of the avatar while a task was in progress.

After learning the task, the instructions and the setup were completed. The above process took approximately 10 minutes. Participants then proceeded to the main part of the experiment. The participants were informed that the main tests were about to begin, and that they would perform ten sets of six tasks, each set followed by a questionnaire (answered in VR), and that it would take between 12 and 20 minutes (measured during the pilots and continued to be accurate). The self-avatar was not visible during the questionnaire. Afterward, they filled out a post-questionnaire (not in VR) about their experience of controlling the self-avatar to complete the tasks, see Table 3.5. In total, the experiment took between 35 and 45 minutes per participant.

### 3.2.5 Measurements

Four performance measurements and two qualitative measurements were made per test condition. Please refer to Section 3.1.5 for additional details on how we collected the performance measurements in the prototype application.

#### Performance measurements per condition:

- *Time* ( $t$ ) is the duration in seconds between when the participant grabs the pickup, and when the pickup reaches the target. Adjusted for induced delay. Average over all tasks in a test.
- *Collision count* is the number of collisions between the participant’s virtual arms and obstacles during the task. Average over all tasks in a test.
- *Collision volume* ( $v$ ) is the volume in  $\text{cm}^3$  of collisions during the task. Each obstacle is divided into smaller parts of  $27 \text{ cm}^3$  each that detect collisions independently (thus, this is the resolution of the collision volume). A collision between such an obstacle part and any of the left or right upper arm, forearm, or hand, adds  $27 \text{ cm}^3$  to the collision volume for the current task. Average over all tasks in a test.
- *Collision duration* ( $d$ ) is the total duration in seconds of all collisions during the task. The duration of a collision between an arm and an obstacle is the duration the arm is located inside of the obstacle. Average over all tasks in a test.

#### Qualitative measurements per condition:

The questionnaire (Q, Table 3.3) on the SoA and SoBO that participants fill out at the end of each test was adapted from the standard questionnaire Virtual Embodiment Questionnaire (VEQ) in [36]. The VEQ consists of three components, one for agency, one for body ownership, and one for change. The change component, which measures the extent one feels the virtual body’s schema was changed, was left out since the virtual avatar’s size, appearance, and body composition are kept consistent throughout the experiment (except resizing the arms for each participant). Each statement in the questionnaire is scored on a 1-7 Likert scale. The scoring for SoA is the average score of the agency statements, and the scoring for SoBO is the average of the body ownership statements. The questionnaire instructions presented to participants were the same ones as in [36]: “Please read each statement and answer on a 1 to 7 scale indicating how much each statement applied to you during the experiment. There are no right or wrong answers. Please answer spontaneously and intuitively. Scale example: 1-strongly disagree. 4-neither agree nor disagree. 7-strongly agree.”. Since the SoA is the main focus of this study we included more statements for that and less for the SoBO. We wanted to keep the number of statements low since this questionnaire was answered for each condition.

Table 3.3. Displays the questionnaire (Q) items on sense of agency (SoA) and sense of body ownership (SoBO). The individual statements are scored on a 1-7 Likert scale (1-strongly disagree, 4-neither agree nor disagree, 7-strongly agree). Based on [36].

| <b>SoA</b>  | <b>Scoring: (AG1 + AG2 + AG3 + AG4 + AG5) / 5</b>                     |
|-------------|-----------------------------------------------------------------------|
| AG1         | The movements of the virtual arms felt like they were my movements.   |
| AG2         | I felt like I was controlling the movements of the virtual arms.      |
| AG3         | I felt like I was causing the movements of the virtual arms.          |
| AG4         | The movements of the virtual arms were in sync with my own movements. |
| AG5         | I enjoyed controlling the virtual arms.                               |
| <b>SoBO</b> | <b>Scoring: (OW1 + OW2) / 2</b>                                       |
| OW1         | It felt like the virtual upper body belonged to me.                   |
| OW2         | It felt like the virtual upper body was my body.                      |

#### Pre- and post-questionnaires:

Aside from the measurements that were made per conditions, participants also filled out a pre-questionnaire (before the main experiment) on immersive tendencies based on [50] seen in Table 3.4, and a post-questionnaire (after the main experiment) on questions related to how it felt to control the self-avatar to complete the tasks when the synchrony was both bad and good, related to the SoA, seen in Table 3.5.

Table 3.4. Displays the questionnaire items in the sparse Immersive Tendencies Questionnaire (sITQ). Based on [50]. The individual statements are scored on a 1-7 Likert scale (1-strongly disagree, 4-neither agree nor disagree, 7-strongly agree). Instructions given to the participants: "Please consider the entire scale when making your responses, as the intermediate levels may apply. For example, if your response is once or twice, the second box from the left should be marked. If your response is many times but not extremely often, then the sixth (or the second box from the right) should be marked. You are not supposed to spend more than a few seconds on each statement."

| <b>Item</b> | <b>Statement / Question</b>                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------|
| IM1         | Do you easily become deeply involved in games, movies or tv dramas?                                         |
| IM2         | How mentally alert do you feel at the present time?                                                         |
| IM3         | How physically fit do you feel today?                                                                       |
| IM4         | How good are you at blocking out external distractions when you are involved in something?                  |
| IM5         | How well do you concentrate on enjoyable activities?                                                        |
| IM6         | How often do you play video games? (often should be taken to mean every day or every two days, on average.) |
| IM7         | Do you ever become so involved in doing something that you lose all track of time?                          |

Table 3.5. Displays the questionnaire items in the post-questionnaire (PQ). Instructions given to the participants: Please read each statement or question carefully and answer honestly. There are no right or wrong answers. How you should interpret "less synchronous" and "more synchronous" in the questions/statements below. Less synchronous = Think about how it felt when you could clearly notice that the virtual arm movements were disturbed compared to your real arm movements (e.g. the worst case). More synchronous = Think about how it felt when you did not perceive any notable (or as little as possible) disturbance in the virtual arm movements compared to your real arm movements.

| Item | Statement / Question                                                                                                                                                                                                                                                                                                            | Options                                                                                                                                                                              |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PQ1  | Did you experience that the movement synchrony between your real and virtual arms were disturbed sometimes during the experiment?                                                                                                                                                                                               | Yes / No                                                                                                                                                                             |
| PQ2  | If yes, what felt off about the virtual arm movements? Try using a few describing words. (Max 200 characters)                                                                                                                                                                                                                   | Short text answer                                                                                                                                                                    |
| PQ3  | Whenever the virtual arm movements felt less synchronous, was it harder to focus on the task?                                                                                                                                                                                                                                   | Yes / No                                                                                                                                                                             |
| PQ4  | Whenever the virtual arm movements felt less synchronous, did you have to put more mental effort into performing the task?                                                                                                                                                                                                      | Yes / No                                                                                                                                                                             |
| PQ5  | Whenever the virtual arm movements felt more synchronous, did you feel more motivated to perform the task?                                                                                                                                                                                                                      | Yes / No                                                                                                                                                                             |
| PQ6  | Whenever the virtual arm movements felt less synchronous, did you pay more attention to the movement of your virtual arms than the obstacles?                                                                                                                                                                                   | Yes / No                                                                                                                                                                             |
| PQ7  | Whenever the virtual arm movements felt more synchronous, did you pay more attention to the movement of your virtual arms than the obstacles?                                                                                                                                                                                   | Yes / No                                                                                                                                                                             |
| PQ8  | Whenever the virtual arm movements felt less synchronous, did the virtual upper body feel more like an external tool to complete the task than a body that belonged to you?                                                                                                                                                     | Yes / No                                                                                                                                                                             |
| PQ9  | Between these three options, what was your main way of noticing that the virtual arm movements felt less synchronous?                                                                                                                                                                                                           | (1) I noticed that the virtual arms were not synchronous to my real arms, (2) I noticed that the task got harder and I performed worse, (3) Somewhat equal mix of both of the above. |
| PQ10 | If you answered (2) in the question above, i.e. that you noticed the task got harder and you performed worse. Do you think that made you more aware of the fact that the virtual arms were under worse control?                                                                                                                 | Yes / No                                                                                                                                                                             |
| PQ11 | Did you feel that some part of the task made it more apparent that the virtual arms were less synchronous?                                                                                                                                                                                                                      | (1) Moving your virtual arm(s) toward and picking up the pickup, (2) Navigating the pickup to the goal while avoiding obstacles, (3) No part made it more apparent than the other.   |
| PQ12 | Whenever the virtual arm movements felt less synchronous, did your movement strategy become more exploitative? That is, did you care less for collisions and opted to move for instance in straight fast movements towards the goal (through the gap)? Or did it become less exploitative? E.g. slower, more planned movements? | (1) More exploitative, (2) Less exploitative, (3) Neither more or less exploitative.                                                                                                 |
| PQ13 | If your strategy to perform the task changed in any way, except becoming more or less exploitative, depending on the synchrony between your real arms and the virtual arms, how did it change?                                                                                                                                  | Text answer                                                                                                                                                                          |
| PQ14 | Open comments about the movements of the avatar                                                                                                                                                                                                                                                                                 | Text answer                                                                                                                                                                          |
| PQ15 | Open comments about the task                                                                                                                                                                                                                                                                                                    | Text answer                                                                                                                                                                          |
| PQ16 | Other comments                                                                                                                                                                                                                                                                                                                  | Text answer                                                                                                                                                                          |

## 4 Evaluation

In this section we present the results of our user study, and evaluate our research questions.

Table 4.1. Displays the results of significance tests and post-hoc tests for the sense of agency (SoA), sense of body ownership (SoBO), and task performance metrics. The first column shows the independent variable and evaluation metric, the second column shows the significance test with its result, and the third column shows the post-hoc test with its result. For the questionnaire (Q), higher values are better. For task performance, higher values mean worse performance. Repeated measures is denoted with "rm". Significance level = .05. All post-hocs use Bonferroni adjustment.

| Evaluation                          | Test                                               | Post-hoc                                                                                                                                                                           |
|-------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Delay vs. Questionnaire (Q)</b>  |                                                    |                                                                                                                                                                                    |
| SoA                                 | One-way rm ANOVA<br>$F_{3,63} = 33.624, p < .0001$ | rm t-tests<br>B > D98 ( $p < .001$ ), B > D165 ( $p < .0001$ ),<br>B > D243 ( $p < .0001$ ), D98 > D165 ( $p = .0017$ ),<br>D98 > D243 ( $p < .0001$ ), D165 > D243 ( $p = .012$ ) |
| SoBO                                | One-way rm ANOVA<br>$F_{3,63} = 10.090, p < .0001$ | rm t-tests<br>B > D165 ( $p = .038$ ), B > D243 ( $p = .0016$ ),<br>D98 > D243 ( $p = .0082$ )                                                                                     |
| <b>Delay vs. Task performance</b>   |                                                    |                                                                                                                                                                                    |
| Time ( $t$ )                        | Friedman test<br>$\chi^2(3) = 11.727, p = .0084$   | Wilcoxon tests<br>B < D165 ( $p = .031$ ), B < D243 ( $p = .012$ )                                                                                                                 |
| Collision volume ( $v$ )            | Friedman test<br>$\chi^2(3) = 25.610, p < .0001$   | Wilcoxon tests<br>B < D165 ( $p = .0039$ ), B < D243 ( $p < .0001$ ),<br>D98 < D243 ( $p < .001$ ), D165 < D243 ( $p = .0038$ )                                                    |
| Collision count ( $c$ )             | One-way rm ANOVA<br>$F_{3,63} = 13.679, p < .0001$ | rm t-tests<br>B < D165 ( $p = .014$ ), B < D243 ( $p < .0001$ ),<br>D98 < D243 ( $p < .001$ )                                                                                      |
| Collision duration ( $d$ )          | Friedman test<br>$\chi^2(3) = 30.818, p < .0001$   | Wilcoxon tests<br>B < D165 ( $p = .0015$ ), B < D243 ( $p < .0001$ ),<br>D98 < D243 ( $p = .0021$ )                                                                                |
| <b>Offset vs. Questionnaire (Q)</b> |                                                    |                                                                                                                                                                                    |
| SoA                                 | One-way rm ANOVA<br>$F_{3,63} = 23.715, p < .0001$ | rm t-tests<br>B > O12 ( $p < .001$ ), B > O20 ( $p < .0001$ ),<br>O06 > O20 ( $p < .0001$ ), O12 > O20 ( $p = .010$ )                                                              |
| SoBO                                | One-way rm ANOVA<br>$F_{3,63} = 7.033, p < .001$   | rm t-tests<br>B > O20 ( $p = .0068$ ), O06 > O20 ( $p = .018$ )                                                                                                                    |
| <b>Offset vs. Task performance</b>  |                                                    |                                                                                                                                                                                    |
| Time ( $t$ )                        | Friedman test<br>$\chi^2(3) = 8.891, p = .031$     | Wilcoxon tests<br>non-significant                                                                                                                                                  |
| Collision volume ( $v$ )            | Friedman test<br>$\chi^2(3) = 9.655, p = .022$     | Wilcoxon tests<br>B < O20 ( $p = .0086$ )                                                                                                                                          |
| Collision count ( $c$ )             | Friedman test<br>$\chi^2(3) = 13.306, p = .0040$   | Wilcoxon tests<br>B < O20 ( $p = .0017$ )                                                                                                                                          |
| Collision duration ( $d$ )          | Friedman test<br>$\chi^2(3) = 14.455, p = .0023$   | Wilcoxon tests<br>B < O20 ( $p < .001$ ), O12 < O20 ( $p = .0056$ )                                                                                                                |
| <b>Mixed vs. Questionnaire (Q)</b>  |                                                    |                                                                                                                                                                                    |
| SoA                                 | One-way rm ANOVA<br>$F_{3,63} = 46.524, p < .0001$ | rm t-tests<br>B > M1 ( $p < .0001$ ), B > M2 ( $p < .0001$ ),<br>B > M3 ( $p < .0001$ ), M1 > M3 ( $p < .0001$ ),<br>M2 > M3 ( $p < .0001$ )                                       |
| SoBO                                | Friedman test<br>$\chi^2(3) = 36.581, p < .0001$   | Wilcoxon tests<br>B > M1 ( $p = .027$ ), B > M3 ( $p < .001$ ),<br>M1 > M3 ( $p < .001$ ), M2 > M3 ( $p = .0046$ )                                                                 |
| <b>Mixed vs. Task performance</b>   |                                                    |                                                                                                                                                                                    |
| Time ( $t$ )                        | Friedman test<br>$\chi^2(3) = 28.745, p < .0001$   | Wilcoxon tests<br>B < M3 ( $p < .0001$ ), M1 < M3 ( $p < .0001$ ),<br>M2 < M3 ( $p = .0050$ )                                                                                      |
| Collision volume ( $v$ )            | Friedman test<br>$\chi^2(3) = 29.603, p < .0001$   | Wilcoxon tests<br>B < M1 ( $p = .0024$ ), B < M2 ( $p = .015$ ),<br>B < M3 ( $p < .0001$ ), M1 < M3 ( $p < .001$ ),<br>M2 < M3 ( $p = .0021$ )                                     |
| Collision count ( $c$ )             | Friedman test<br>$\chi^2(3) = 33.827, p < .0001$   | Wilcoxon tests<br>B < M1 ( $p = .0066$ ), B < M2 ( $p = .0055$ ),<br>B < M3 ( $p < .0001$ ), M1 < M3 ( $p = .0074$ ),<br>M2 < M3 ( $p = .012$ )                                    |
| Collision duration ( $d$ )          | Friedman test<br>$\chi^2(3) = 36.927, p < .0001$   | Wilcoxon tests<br>B < M1 ( $p = .0015$ ), B < M2 ( $p = .0050$ ),<br>B < M3 ( $p < .0001$ ), M1 < M3 ( $p < .001$ ),<br>M2 < M3 ( $p > .0001$ )                                    |

## 4.1 Results

In this section, we display the results of the user study described in Section 3.2. For details on experimental conditions, see Section 3.2.1, and for details on the measurements see Section 3.2.5. Table 4.1 lists the results of all significance tests and post-hocs.

Significance testing was done to evaluate the effect of delay, offset, and combined (mixed) on each qualitative and quantitative metric. For each such evaluation (e.g. SoA vs. Delay), we checked for normality using D’Agostino and Pearson test [10] and sphericity using Mauchly’s sphericity test (both significance level  $\alpha = .05$ ). If the data could be assumed to come from normal distributions and to be spherical, we performed One-way repeated measures ANOVA for the significance test. Followed by pairwise repeated measures t-tests for each condition within the category (e.g. for delay, pairwise between conditions B, D98, D165, D243). Otherwise, we performed the Friedman test (nonparametric alternative to One-way repeated measures ANOVA). Followed by pairwise Wilcoxon tests (nonparametric alternative to repeated measures t-test). We use significance level = .05, and all post-hoc p-values are adjusted with Bonferroni adjustment. Please refer to Table 4.1 for details on the significance tests. Mixed condition values for reference, M1: (98 ms, 12 cm), M2: (165 ms, 6 cm), M3: (243 ms, 20 cm). Milliseconds (ms) and centimeters (cm).

### 4.1.1 Sense of agency and body ownership

The questionnaire (Q) results on sense of agency (SoA) and sense of body ownership (SoBO) are presented first. Mean and standard deviation values are summarized in Table 4.2. Boxplots of the SoA and SoBO are shown in Figure 4.1 and Figure 4.2, respectively.

Table 4.2. Displays the mean and standard deviation (in parenthesis) of sense of agency (SoA) and sense of body ownership (SoBO) for each delay, offset, and mixed condition. Higher mean is better.

|               | SoA        | SoBO       |
|---------------|------------|------------|
| B             | 5.91(1.03) | 4.91(1.80) |
| <b>Delay</b>  |            |            |
| D98           | 5.26(1.16) | 4.41(1.85) |
| D165          | 4.38(1.41) | 3.91(1.72) |
| D243          | 3.33(1.20) | 3.14(1.75) |
| <b>Offset</b> |            |            |
| O06           | 5.45(0.96) | 4.70(1.32) |
| O12           | 5.15(1.09) | 4.11(1.79) |
| O20           | 4.30(1.33) | 3.66(2.05) |
| <b>Mixed</b>  |            |            |
| M1            | 4.56(1.42) | 3.86(2.14) |
| M2            | 4.43(1.22) | 3.70(1.86) |
| M3            | 2.54(1.05) | 2.05(1.65) |

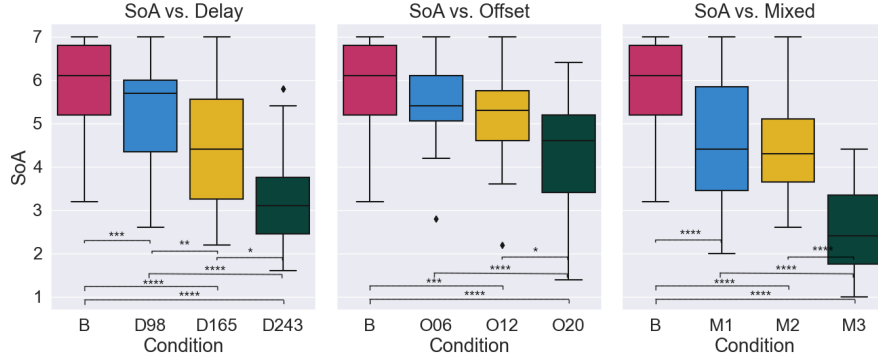


Figure 4.1. Displays boxplots of the sense of agency (SoA) for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

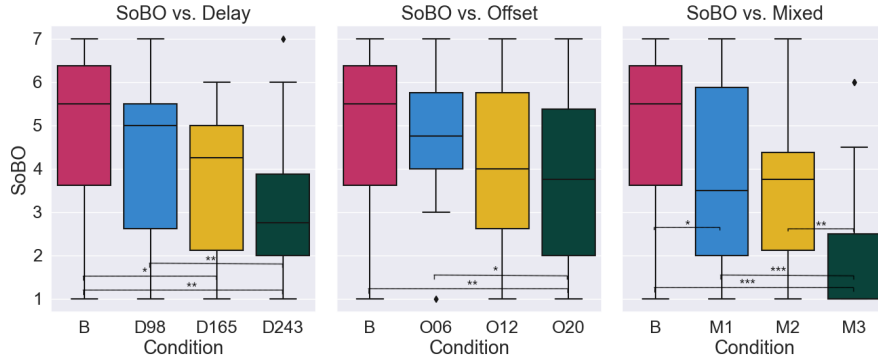


Figure 4.2. Displays boxplots of the sense of body ownership (SoBO) for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

User reported SoA was significantly affected by delay conditions, offset conditions, and mixed conditions (see Table 4.1). Figure 4.1 shows that higher delays result in lower reported SoA. Post-hocs showed that there is a significant negative effect between all delay conditions. Notably, that includes between baseline B and D98. Offset did not have as large of a negative effect on the SoA as motion delay with our conditions. Post-hocs showed that a significant negative effect was first experienced between B and O12. No reduction was found between B and O06. Reductions in SoA were also found in between conditions in the following pairs of offset conditions: (B, O20), (O06, O20), and (O12, O20). Figure 4.1 shows that SoA in general decreases with increases in

offset, but not as drastically between conditions as with delay. This is most evident when comparing their lower quartile values for SoA. Note the difference in SoA between D165 and D243. When combining delay and offset in the mixed conditions, we see a heavily decreased SoA between B and M3. Post-hoc analysis shows that a significant decrease in SoA was experienced between all mixed conditions except M1 and M2. Note that M1 has a 98 ms delay and 12 cm offset, while M2 has a 165 ms delay and 6 cm offset, therefore it is not a decrease in both delay and offset between them. Looking at the SoA column in Table 4.2 we can compare the means of all synchrony conditions. We note that baseline SoA is 5.91, the worst delay condition D243 resulted in 3.33, the worst offset condition O20 in 4.3, and the worst condition overall M3 in 2.54. The closest value to baseline is for O06.

Similarly for SoBO, it was significantly affected by delay conditions, offset conditions, as well as mixed conditions (see Table 4.1). Figure 4.2 shows that higher delays in general led to lower reported SoBO, this is most evident when looking at the median and upper quartile. Post-hocs showed delay had a significant negative effect on SoBO first between B and D165. Negative effects were also found between conditions in the following pairs: (B, D243) and (D98, D243). Increasing offset generally led to lower SoBO, see median values in Figure 4.2. However, post-hocs showed that there is only a significant negative reduction between B and O20, as well as O06 and O20. For the mixed conditions we want to note that for M3, reported SoBO completely breaks down, with a median of 1. Post-hocs showed significant negative effects between conditions in the following pairs: (B, M1), (B, M3), (M1, M3), and (M2, M3). Note that no significant decrease was found between B and M2. Again, looking at SoBO column in Table 4.2 we can compare the means of all synchrony conditions. Baseline SoBO is 4.91, D243 resulted in 3.14, O20 in 3.66, and M3 in 2.05. The closest value to baseline is for O06.

Further post-hoc analysis ( $\alpha = .05$ ) was made to compare each mixed condition to their closely related offset and delay conditions. For instance, we compared M3 (243 ms delay, 20 cm offset) to both D243 and O20 to see if we could find significant differences when adding e.g. an offset to an already bad synchrony. Note that these are not included in Table 4.1. To test the effect on SoA we used pairwise repeated measures t-tests with Bonferroni adjustment and got the following results: (M1 < D98,  $p = .016$ ), (M2 < O06,  $p < .001$ ), (M3 < D243,  $p = .0085$ ), (M3 < O20,  $p < .0001$ ). Therefore regarding SoA, adding an offset of 12 cm to a delay of 98 ms has a significant effect, adding a 165 ms delay to an offset of 6 cm has a significant effect, adding a 20 cm offset to a 243 ms delay has significant effect, and adding a 243 ms delay to a 20 cm offset has significant effect. For SoBO, since data for some of the mixed conditions violated the normality assumption, we used pairwise Wilcoxon tests with Bonferroni adjustment and found the following: (M2 < O06,  $p = .041$ ), (M3 < D243,  $p = .0011$ ), (M3 < O20,  $p = .0017$ ). Here, adding a 20 cm offset to a 243 ms delay has a significant effect, and adding a 243 ms delay to a 20 cm offset has a significant effect. Note that adding smaller delays and offsets did not see a significant effect on SoBO.

Finally, we analyzed the relationship between delay and offset on both SoA and SoBO. We assumed that the relationship is linear and performed Ordinary Linear Regression (OLS) using data from all synchrony conditions. Let  $x$  represent delay, and  $y$  represent offset. For SoA we found the following relationship:  $\text{SoA} = -0.0104x + -0.0062y + 6.1845$  (adjusted  $R^2 = 0.379$ ,  $F = 67.88$ ,  $Pr(F) < .0001$ ). Offset was supplied in millimeters for a scale similar to the delay. The OLS assumption of normally distributed errors was violated for SoBO (and with a low adjusted  $R^2 = 0.152$ ). Still, this was the regression result:  $\text{SoBO} = -0.0080x + -0.0061y + 5.2185$ . We also performed Pearson correlation between reported SoA and SoBO across all synchrony conditions and found a fairly strong correlation ( $r = 0.754$ ,  $p < .0001$ ).

#### 4.1.2 Task performance

The task performance results are summarized in Table 4.3 showing means and standard deviation for each synchrony condition. Boxplots of each metric are shown in Figures 4.3, 4.4, 4.5, and 4.6. The performance measurements had some outliers. However, we had no reason to suspect these measurements, and removing them would make our experimental design unbalanced.

Table 4.3. Displays the mean and standard deviation (in parenthesis) of the task performance metrics (time  $t$ , collision volume  $v$ , collision count  $c$ , and collision duration  $d$ ) for each delay, offset, and mixed condition. Higher mean is worse task performance.

|               | $t$ (s)    | $v$ (cm <sup>3</sup> ) | $c$        | $d$ (s)    |
|---------------|------------|------------------------|------------|------------|
| B             | 3.77(1.88) | 499.9(375.8)           | 1.72(0.80) | 0.33(0.20) |
| <b>Delay</b>  |            |                        |            |            |
| D98           | 4.25(1.86) | 641.7(689.2)           | 2.02(0.97) | 0.47(0.44) |
| D165          | 4.54(2.74) | 819.2(550.5)           | 2.45(1.10) | 0.70(0.48) |
| D243          | 4.69(2.37) | 1342.4(1083.4)         | 3.02(1.67) | 0.87(0.45) |
| <b>Offset</b> |            |                        |            |            |
| O06           | 3.89(1.96) | 609.8(566.7)           | 2.08(1.18) | 0.49(0.37) |
| O12           | 3.93(2.07) | 902.9(1280.3)          | 2.10(1.34) | 0.48(0.35) |
| O20           | 4.15(1.79) | 971.4(1003.7)          | 2.54(1.47) | 0.77(0.63) |
| <b>Mixed</b>  |            |                        |            |            |
| M1            | 3.79(1.78) | 896.7(725.3)           | 2.44(1.30) | 0.68(0.57) |
| M2            | 4.06(1.94) | 1045.8(877.8)          | 2.52(1.02) | 0.63(0.34) |
| M3            | 5.35(2.23) | 1947.1(1438.1)         | 3.39(1.45) | 1.67(1.50) |

As seen in Table 4.1, all task performance metrics (time, collision volume, collision count, and collision duration) were significantly affected by the delay conditions, the offset conditions, and the mixed conditions.

Figure 4.3 shows that task time did not see an evident increase as delay was increased. However, post-hocs revealed that a significant increase exists between B and D165, as well as between B and D243. Note that the measured time was corrected by subtracting the induced delay. No significant effect of increasing offset was found on task time by the post-hocs. For the mixed conditions, the

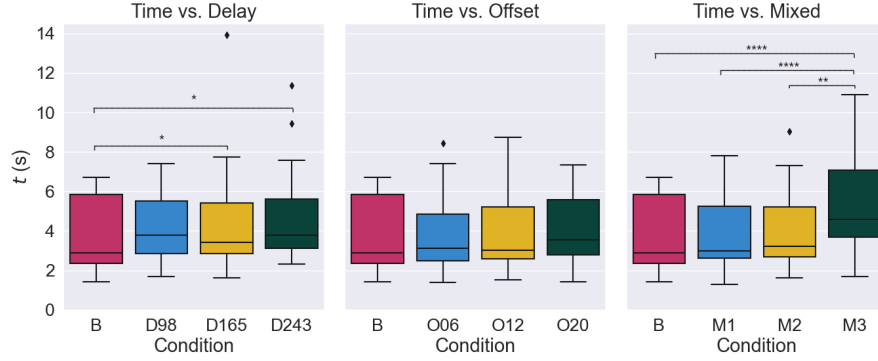


Figure 4.3. Displays boxplots of the time  $t$  (s) performance metric for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

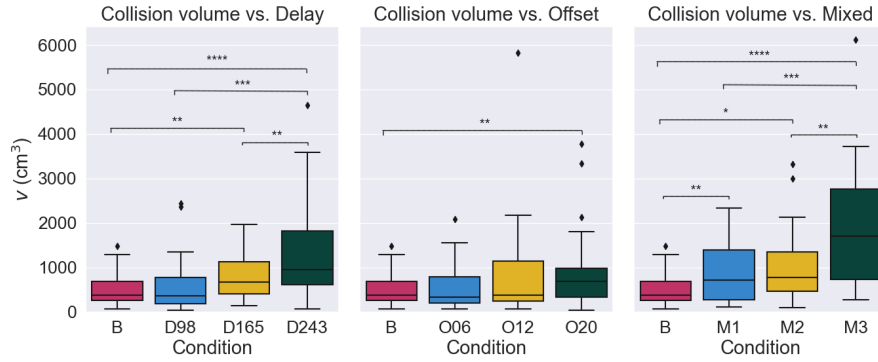


Figure 4.4. Displays boxplots of the collision volume  $v$  ( $\text{cm}^3$ ) performance metric for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

post-hocs showed a significant increase in time between the conditions in pairs (B, M3), (M1, M3), and (M2, M3). Looking at the values in the time column in Table 4.3, delay conditions show the highest mean task times, except M3 which shows the highest mean task time of 5.35 seconds. Baseline mean task time was 3.77 seconds.

Figure 4.4 shows the effect of the synchrony conditions on task collision volume. Collision volume increased as delay increased, with post hocs showing a significant increase first between B and D165. But also between conditions in pairs (B, D243), (D98, D243), and (D165, D243). Offset did not affect collision volume as much, with only one significant post-hoc between B and O20. Among

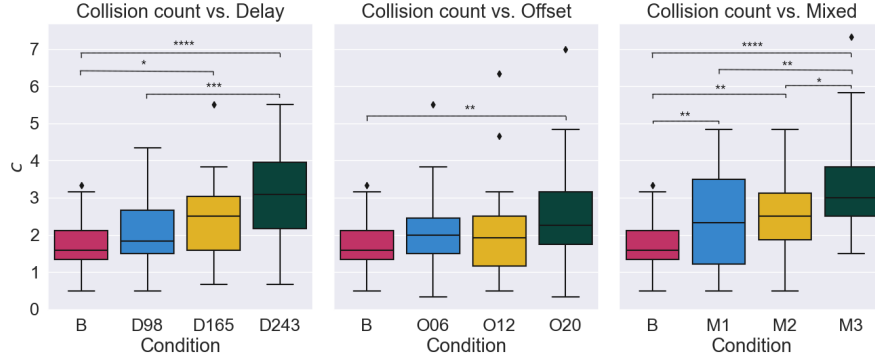


Figure 4.5. Displays boxplots of the collision count  $c$  performance metric for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

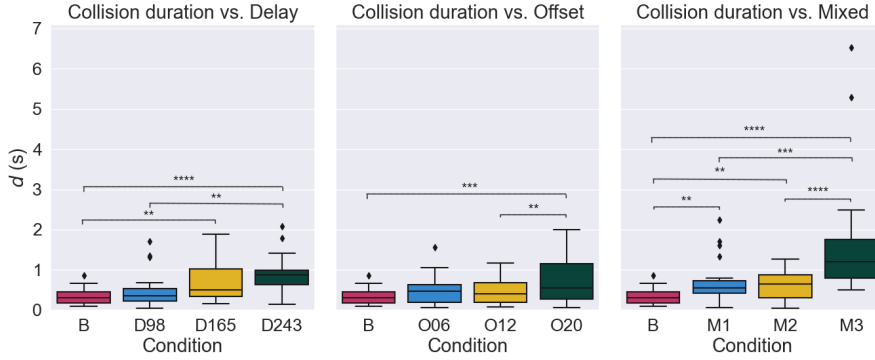


Figure 4.6. Displays boxplots of the collision duration  $d$  (s) performance metric for each synchrony condition in delays (left), offsets (middle), and mixed (right). Significance levels between conditions are represented by asterisks and the corresponding connecting line: \* ( $p < .05$ ), \*\* ( $p < .01$ ), \*\*\* ( $p < .001$ ), \*\*\*\* ( $p < .0001$ ).

the mixed conditions, a significant effect was found between all conditions except between M1 and M2. However, in the collision volume column of Table 4.3 we see that the mean values steadily increase as the synchrony gets worse. It is not as easy to see in Figure 4.4 because of the y-axis scale. The baseline condition shows a mean task collision volume of  $499.9 \text{ cm}^3$ , the worst delay condition D243 shows  $1342.2 \text{ cm}^3$ , the worst offset condition O20 shows  $971.4 \text{ cm}^3$ , and the worst overall condition M3 shows  $1947.1 \text{ cm}^3$ .

Task collision count increased with increasing delay and offset, but again more noticeably with delay, as shown in Figure 4.5. Post-hocs showed that collision count increased significantly first between delay conditions B and D165.

But also between conditions in pairs (B, D243), and (D98, D243). Offset caused a significant increase in collisions first between B and O20. Collision count was significantly affected between all pairs of mixed conditions except between M1 and M2. Looking at the collision count column of Table 4.3, we find B at a task mean of 1.72 collisions, D243 at 3.02 collisions, O20 at 2.54 collisions, and M3 at 3.39 collisions.

As shown by Figure 4.6, task collision duration also increased with worse synchrony conditions. Post-hocs show that delay causes an increase first between B and D165. But, also between conditions in pairs (B, D243), and (D98, D243). Offset was shown to be significant first between B and O20, but also between O12 and O20. Notably, not between O06 and O20. Conditions O06 and O12 resulted in a very similar distribution of collision durations (see Table 4.3 as well). In the collision duration column of Table 4.3 we find that B resulted in a mean of 0.33 seconds task collision duration, D243 in 0.87 seconds, O20 in 0.77 seconds, and M3 drastically higher at 1.67 seconds (note the two outliers shown in Figure 4.6).

As with SoBO and SoA, we also examined whether or not the mixed conditions resulted in significantly reduced task performance compared to their similar delay and offset conditions. Note that these post-hocs are not included in Table 4.1. This was done using pairwise Wilcoxon tests with Bonferroni adjustment since all evaluations contained at least one group that violated the normality assumption. For task time we found ( $M3 > D243$ ,  $p = .047$ ), ( $M3 > O20$ ,  $p < .001$ ). For task collision volume we found ( $M2 > O06$ ,  $p = .034$ ) ( $M3 > D243$ ,  $p = .025$ ), ( $M3 > O20$ ,  $p < .001$ ). For task collision count we found ( $M3 > O20$ ,  $p = .0078$ ). For task collision duration we found ( $M1 > D98$ ,  $p = .031$ ), ( $M3 > D243$ ,  $p = .0050$ ), ( $M3 > O20$ ,  $p < .001$ ).

Linear regression using OLS was performed to explore the relation between delay and offset on the task performance metrics as well. However, all of the results violated the OLS assumption of normally distributed errors, and had bad fits.

In general, the delay conditions had more significant effects on the task performance metrics than the offset conditions, while the worst synchrony condition M3 always had the largest negative impact. Regarding which delays and offsets task performance is first seen affected by compared to baseline. No significantly reduced task performance was found between B and D98, or B and O06. Task performance was usually impacted negatively between B and D165 for delay, and between B and O20 for offset. However, among the mixed conditions, task performance (across all metrics except task time) took a significant hit between B and either M1 or M2.

### 4.1.3 Pre-questionnaires

To examine if immersive tendencies in participants had an effect on their reported SoA and SoBO we used Pearson correlation. See the sITQ questionnaire in Table 3.4. We found no significant correlation between the average sITQ and average SoA in participants. Between questionnaire item 4 "How good are

you at blocking out external distractions when you are involved in something?" and average SoA we found a weak correlation ( $r = 0.369, p = .045$ ). Between questionnaire item 5 "How well do you concentrate on enjoyable activities?" and average SoA we found a moderate correlation ( $r = 0.483, p = .011$ ). We found no significant correlation between the average sITQ and average SoBO in participants. Again, between questionnaire item 4 and average SoBO we found a weak correlation ( $r = 0.373, p = .043$ ). Between item 5 and average SoBO we found another weak correlation ( $r = 0.417, p = .027$ ). No significant correlation was found between VR experience and any of the dependent metrics.

#### 4.1.4 Post-questionnaire

For the post-questionnaire (PQ, see Table 3.5) results we first list the Yes/No percentage for appropriate statements in Table 4.4. Then, the answers to the multiple choice statements. Finally, we present a summary of the textual responses for items where the participants could write freely.

One participant answered "No" on PQ1, but we suspect they noticed something since they asked if there was a technical issue with the arms the first time we reduced the motion synchrony. Unfortunately, we did not get the chance to properly ask them about it since the response was noticed later when we looked at the data. Still, their responses are included in the statistics.

For PQ9 "Between these three options, what was your main way of noticing that the virtual arm movements felt less synchronous? Options: (1) I noticed that the virtual arms were not synchronous to my real arms, (2) I noticed that the task got harder and I performed worse, (3) Somewhat equal mix of both of the above", **68.2 %** of the participants answered (1) and 31.8 % answered (3). No one answered (2). Rendering PQ10 "If you answered (2) in the question above, i.e. that you noticed the task got harder and you performed worse. Do you think that made you more aware of the fact that the virtual arms were under worse control?" useless.

For PQ11 "Did you feel that some part of the task made it more apparent that the virtual arms were less synchronous? Options: (1) Moving your virtual arm(s) toward and picking up the pickup, (2) Navigating the pickup to the goal while avoiding obstacles, (3) No part made it more apparent than the other.", **45.4 %** of participants answered (1), 27.3 % answered (2), and 27.3 % answered (3).

For PQ12 "Whenever the virtual arm movements felt less synchronous, did your movement strategy become more exploitative? That is, did you care less for collisions and opted to move for instance in straight fast movements towards the goal (through the gap)? Or did it become less exploitative? E.g. slower, more planned movements? Options: (1) More exploitative, (2) Less exploitative, (3) Neither more or less exploitative", **50 %** of participants answered (1), 40.9 % answered (2), and 9.1 % answered (3).

Item PQ2: "If yes, what felt off about the virtual arm movements? Try using a few describing words.", where "If yes" refers to PQ1 (see Table 4.4). Commonly seen in answers was "Slow", "Wrong place", "Delayed", "Far apart", and

Table 4.4. Displays the results of the Yes/No statements of the post-questionnaire. First column shows the questionnaire item, second column shows the statement, third column shows the percentage of "yes" answers, and fourth column shows the percentage of "no" answers.

| Item | Statement                                                                                                                                                                   | Yes           | No            |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|
| PQ1  | Did you experience that the movement synchrony between your real and virtual arms were disturbed sometimes during the experiment?                                           | <b>95.5 %</b> | 4.5 %         |
| PQ3  | Whenever the virtual arm movements felt less synchronous, was it harder to focus on the task?                                                                               | <b>72.7 %</b> | 27.3 %        |
| PQ4  | Whenever the virtual arm movements felt less synchronous, did you have to put more mental effort into performing the task?                                                  | <b>95.5 %</b> | 4.5 %         |
| PQ5  | Whenever the virtual arm movements felt more synchronous, did you feel more motivated to perform the task?                                                                  | <b>81.8 %</b> | 18.2 %        |
| PQ6  | Whenever the virtual arm movements felt less synchronous, did you pay more attention to the movement of your virtual arms than the obstacles?                               | <b>90.9 %</b> | 9.1 %         |
| PQ7  | Whenever the virtual arm movements felt more synchronous, did you pay more attention to the movement of your virtual arms than the obstacles?                               | 9.1 %         | <b>90.9 %</b> |
| PQ8  | Whenever the virtual arm movements felt less synchronous, did the virtual upper body feel more like an external tool to complete the task than a body that belonged to you? | <b>81.8 %</b> | 18.2 %        |

"Misaligned", where 76.2 % of answers contained words or phrases semantically similar to "Delay", and 71.4 % of answers contained words or phrases semantically similar to "Offset". Other phrases were for instance "Didn't sync", "Not in sync". Notably, two participants wrote that the elbow positions didn't match their real elbows, one participant wrote that torso orientation was slightly off, and one participant wrote that the wrists bent too much in certain poses. One interesting response was that when the arms were far apart, it was described as feeling like someone standing behind them and mimicking their arm movements.

Item PQ13: "If your strategy to perform the task changed in any way, except becoming more or less exploitative, depending on the synchrony between your real arms and the virtual arms, how did it change?". Out of the 22 participants, 9 people answered this question, and 4 of those answers were filtered out because the answer was similar to their response to PQ12, PQ6, or PQ7. In the 5 remaining answers, strategies were, paraphrased: "focused more on angling my

body correctly”, ”thought of it as an exoskeleton”, ”moved slower”, and ”more focus on timing”.

Item PQ14: ”Open comments about the movements of the avatar” was answered by 12 out of the 22 participants. Three participants noted that the elbow or bend direction of the arms was not always correct. Two people noted that it felt like the arms came from someone behind them when the arms were far apart. One participant mentioned that they felt slightly frustrated whenever the arms were too slow, but that a slight delay still felt immersive. One participant noted that they did not think too much of the avatar. Two participants noted that the avatar felt good in general. One participant wrote that close-to-body motions felt weird when the arms were spaced far apart and that the virtual arms felt clumsy.

Item PQ15: ”Open comments about the task” was answered by 7 out of the 22 participants. Four participants wrote that the task was ”good”, ”fun”, or ”fine”. Two participants noted that it was hard to avoid the obstacles, one of them explained they got more careless when they understood that the estimated arm pose was not always correct. One participant was surprised by the level of immersion they felt when the arms were synchronized, and that going from high synchrony to low synchrony felt very jarring.

Finally, we list some spoken comments from participants during the experiments. One participant commented that the arms felt okay even when offset, but that delay substantially reduced their enjoyment. The same participant missed the mouse when trying to grab it to answer the post-questionnaire and commented about it. Another participant commented that they were surprised that their real arms felt weird afterward, out of sync, too small, being positioned wrong, and did not move like they expected. One participant commented that they tried to look at their watch while they were in VR.

## 4.2 Discussion

Having and controlling a virtual self-avatar is an important contributor to feeling present in VR [39, 11, 15]. Maintaining synchrony, both temporal and positional, between the user’s real motion and the motion of the self-avatar is critical to delivering a strong sense of agency (SoA). Since the feeling of agency arises from the process of intention, action, and effect, disruptions in this chain can negatively affect the SoA [49]. In general, adding a delay between user action and the feedback of the action diminishes the SoA in most action-effect scenarios depending on the length of the delay [48]. In regards to self-avatars in VR, motion delays of approximately 125-150 ms (milliseconds) have been shown to reduce the SoA over the avatar in tasks that don’t require interaction with objects in the virtual environment [46, 28]. The impact of positional offsets on the SoA has not been as widely examined as delay. One study found that adding an offset that corrected users hand position in a VR tracing task, thereby increasing their perceived task performance, had a positive result on the SoA, even if the motion became less accurate to their real motion [1]. Offsets have also been found to affect SoA when they cause mismatches in self-contact [6].

Besides these results, the effect of offsets in VR self-avatars has seen little prior interest.

The main purpose of this study was to examine the impact of disturbed synchrony via delay and offset, between the user’s real and virtual arm motion, on the SoA, in a task where the user has to move their virtual self-avatar’s arms to interact with the virtual environment. To do this, we developed an application prototype containing an upper body self-avatar animated using inverse kinematics, functionality to induce delay and offset on its arm motion, and a task scenario where the user picks up objects and navigates them through obstacles, inspired by [52]. We constructed a repeated measures user study where the participants experienced several synchrony conditions with different delays and offsets. For each condition, we measured the SoA and sense of body ownership (SoBO) via questionnaire, and task performance directly in the application prototype. Task performance was measured since it has been shown to affect and work as an indicator of the SoA [49, 46]. Body ownership was measured for comparison with similar studies that measure the SoBO in addition to the SoA, e.g. in [28, 46]. Participants in the user study also answered a pre-questionnaire on immersive tendencies and a post-questionnaire related to the SoA.

Our general research question (GRQ, Section 1.1) reads ”How does disturbed synchrony between real and virtual arm motions in virtual reality self-avatars affect the sense of agency?” This question does not have a clear-cut answer. Depending on the task scenario, different synchrony conditions have differently sized effects on the SoA, or no effect at all (e.g. [28]). In our task scenario, based on our results and using our synchrony conditions, both delay and offset negatively impacted the SoA. This is what we expected since within the comparator model of agency (Section 2.2.2), both delay and offset in self-avatar arm motion can cause discrepancies in sensorimotor feedback when the user moves their real arms. We found that delay impacts the SoA already at 98 ms, and offset impacts it at 12 cm (centimeters). Increases in delay and offset continued to decrease the SoA further. In fact, every increase in delay had a significant effect on the SoA, suggesting that SoA is severely impacted by motion delay in scenarios similar to ours. The worst SoA was found when combining a 243 ms delay with a 20 cm offset, where the reported SoA was 2.54 on average (the best reported SoA was 5.91, in the baseline condition). This was significantly worse than when applying a 243 ms delay, or 20 cm offset, individually. We can conclude that in tasks where users have to manipulate objects in the virtual environment using their self-avatar arms, and where they have to be constantly aware of their arm motions to not collide with obstacles, disturbed synchrony via both delay and offset, even at relatively low values, negatively impact the SoA over the self-avatar arms, and the effect becomes larger when the disruptions become worse. In the context of self-avatars in VR, these findings present a new lower bound for when delay affects the SoA, and is to our knowledge the first time that the impact of delay and offset has been examined in scenarios similar to ours.

In the next Sections 4.2.1, 4.2.2, and 4.2.3, we discuss the individual effects of delay and offset, as well as when they are combined, in the context of previous

literature. In order to answer our main research questions and give additional context to the general one.

#### **4.2.1 MRQ1: Effect of motion delay on the sense of agency and task performance**

Our first main research question reads "What is the effect of increasing the motion delay {B, D98: 98 ms, D165: 165 ms, D243: 243 ms} on the sense of agency and task performance?". Milliseconds (ms) and centimeters (cm). This question warrants a longer discussion since a lot of previous work has been done to investigate the effect of delays on the SoA.

We expected motion delay to affect both SoA and task performance. Based on the comparator model of agency, adding a delay between action and effect alters the actual feedback compared to the expected feedback when the user acts. This is a common way of altering the SoA as discussed by Wen [48]. Unsurprisingly, previous studies have shown that delay negatively affects the SoA in VR self-avatars as long as users are aware of their body motion [28, 36]. Task performance with self-avatars has also been shown to be affected by delay [43, 46]. In line with these previous studies, we also found that a delay between real arm motion and virtual arm motion affects both SoA and task performance.

We expected higher delay would lead to worse SoA and worse task performance based on previous results. Using a cave automatic virtual environment (CAVE) and in a pose-mimicking task, Waltemate et al. [46] found that higher delays resulted in worse task performance and worse SoA. Roth and Latoschik [36] reported that higher delay leads to worse SoA when the user follows movement instructions and observes their motion. We found that higher delays result in worse SoA and task performance in our manipulation task. Compared to our baseline condition with a 32 ms delay, SoA was negatively impacted first at 98 ms. Comparatively, this is lower than what was found in [46] where SoA was first impacted between 45 ms and 125 ms, which is the lowest bound we could find prior to our study. This is surprising since the mirror observation mode used in [46] has been shown to be the most sensitive to motion artifacts [28]. Several factors could explain this. They did not find significance between 45 ms and 75 ms, but this does not mean they would not have found significance at higher values than 75 ms but lower than 125 ms, should they have included them. Their task required the participants to be aware of their whole upper body pose, while ours only required them to be aware of their arm motion. It is reasonable to suggest that focusing on a more limited area of the body might lead to it being easier to notice smaller deviations in that limited area. Also, the mirror observation mode was found to be more sensitive when the user was immersed using a head-mounted display. Using a CAVE to provide the mirror observation mode might not have the same effect on sensitivity since the CAVE does not hide the user's real body.

In our study, reductions in SoA were significant between all our delay conditions. This suggests that delay severely impacts the SoA in scenarios similar to ours. We found that SoA values steadily decreased from 5.91 in the baseline

condition, to 3.33 in the worst delay condition with 243 ms delay, a 43 % reduction. Compared to [46], going from 45 ms (their baseline) to 350 ms, they saw approximately a 30 % reduction. However, they used a different measurement for the SoA. Comparing with [36], they found SoA to be significantly affected between 90 ms and 353 ms, and the reduction was from approximately 6 to 4.5 (25 % reduction). This seems low compared to Koiliias et al. [28] who found a significant reduction (5.5 to 3, 45 % reduction) between their baseline and 150 ms. Note that both [28] and [36] used full-body setups and tasks where the participants followed movement instructions and simply observed the self-avatar’s motion, no precise motor task was involved. Since the movement instructions were not reported it is hard to explain the difference. Based on these comparisons, our results indicate that the SoA over at least the self-avatar’s arms is heavily sensitive to reductions in delay when the user has to constantly be aware of their arm motions to perform a task, in combination with viewing their avatar from a first-person perspective.

We found that task performance in terms of time, collision volume, collision count, and collision duration, was first negatively impacted between baseline and 165 ms. Adding further delay caused the task performance to decrease further. Even though no significant effect was found between baseline and 98 ms, mean values of the collision metrics went up across the board. In general, task performance continued to decrease when delay increased, especially in terms of the collision metrics. Time was not as affected. We observed that some participants slowed down their movements quite drastically at higher delays or used more time to plan their movement path, while others quickly tried to “shoot” their arm through the gaps instead.

Finally, concerning this research question, we expected task performance to be impacted at lower delay values than SoA based on what was found by Waltemate et al. [46], who found performance to be affected at 75 ms in their task. Instead, we found that SoA was impacted at lower delays than task performance (98 ms vs. 165 ms). This could simply be because of the differences between the tasks. It could be argued that the performance in their task of mimicking a moving pose with the upper body is more sensitive to small delays than our task of avoiding obstacles with our arms. In our task participants were mostly concerned with moving one arm, while in their they had to consider the whole upper body. We reasoned that since perceived task performance is known to influence the user’s judgment of the SoA, for example, shown in [46, 1], we would see task performance be impacted before the SoA, which would then influence the users’ reported SoA when they perceived a lower task performance. In the post-questionnaire (PQ) when asked what the main way of noticing the disrupted synchrony was, 68.2 % of participants answered that they noticed the change in how their arms moved and that it didn’t have to do with their ability to perform the task. While the rest reported that it was a combination of task performance and arm motion. This only suggests that perceived task performance was not the main way participants noticed changes in disrupted synchrony, which makes sense since worse synchrony directly affects arm motion, but not necessarily perceived task performance. The fact that the

task only started once the participant grabbed the pickup object made it so the participants could spend a few seconds feeling the arm motion before focusing on the task, which may have contributed to this. We discuss why we believe both arm motion and perceived task performance had an effect in Section 4.2.6.

Based on what we’ve found, we recommend VR experiences that let users embody a self-avatar and aim for a high SoA to stay below motion delays of 98 ms (preferably close to our baseline if possible). For task performance, motion delays should be at most 98 ms. Particularly if there are tasks involved where users need to be focused on their arm motion and interact in the virtual environment. In the future, it could be interesting to investigate slightly lower delays than 98 ms in a similar task to try and find an even lower threshold.

#### **4.2.2 MRQ2: Effect of position offset on the sense of agency and task performance**

Our second main research question reads "What is the effect of increasing the position offset {B, O06: 6 cm, O12: 12 cm, O20: 20 cm} on the sense of agency and task performance?" Milliseconds (ms) and centimeters (cm).

We expected offset to affect both SoA and task performance based on the comparator model of agency. Offset-based artifacts such as popping have been shown to affect the SoA [28]. Evidence also suggests that offsets impact the SoA when self-contact is involved. [6]. However, we did not find any previous studies that examined the impact of offset on SoA in tasks similar to ours. Groen and Werkhoven [18] found that a 10 cm offset did not affect task performance in a manipulation task, but we hypothesized that offset might affect task performance at higher values, for example, 20 cm. The results from our user study show that both SoA and task performance are significantly affected by offset. However, the post-analysis shows that the effect on task performance is not profound.

We expected higher offsets would lead to decreased SoA and task performance since the discrepancy between the user’s real and virtual arm motion becomes larger, and we don’t expect our offsets to increase perceived task performance as in [1]. Our results show that SoA is first affected between baseline and 12 cm offset. With self-contact, offsets of 7.5 cm were significant on SoA in [6], which probably can be explained by the additional expectation of haptic feedback. The SoA steadily decreased from 5.91 at baseline to 4.30 at 20 cm offset. The impact is not as large as with delay within our conditions, but it is significant. Collision metrics were impacted first between baseline and 20 cm offset (our worst offset condition). But not between baseline and 12 cm, which is in line with [18]. The time performance metric was not affected at all. Based on the comparator model of agency, we suggest the lesser impact could be because the internal model for motor control is more easily adapted to static offsets in arm position than it is to delays between real and virtual arm motion. This leads to fewer prediction errors about the virtual arm motion, and subsequently, both the SoA and the task performance are not affected as much.

In terms of the task, we observed that participants usually tried to compensate for the offset by angling or adjusting their body’s position.

We suggest that offsets between the user’s and the self-avatar’s arms should be at most 6 cm for it to not impact the SoA, and at most 12 cm to not affect task performance. Again, particularly if tasks are involved that require the user to focus on their arm motion and interact in the virtual environment.

### **4.2.3 MRQ3: Effect of combined delay and offset on the sense of agency and task performance**

Our third main research questions reads ”What is the effect of combining both delay and offset {B, M1: (98 ms, 12 cm), M2: (165 ms, 6 cm), M3: (243 ms, 20 cm)} on the sense of agency and task performance?” Milliseconds (ms) and centimeters (cm).

For this research question, we expected that combining delay and offset would result in worse SoA and task performance compared to the same condition when either delay or offset is omitted. E.g. (98 ms, 12 cm) was expected to be worse than both (98 ms, 0 cm) and (32 ms, 12 cm). Based on the fact that there will be a larger sensorimotor feedback mismatch, again explained via the comparator model of agency. For SoA, we found this to be true for the condition with 243 ms delay and 20 cm offset, compared to the individual 243 ms delay, and 20 cm offset conditions. All in all, combining the worst delay with the worst offset yielded the worst SoA score across all conditions of 2.54. We also found that adding a 12 cm offset to a 98 ms delay gave a significant decrease. So did adding 165 ms delay to 6 cm offset. Comparing the impact of delay and offset, our linear regression analysis of SoA suggested, though slightly unreliably, that across all conditions, delay (milliseconds) has almost double the impact on SoA compared to offset (millimeters). We reason why offset has a lower impact in the previous section. As for task performance, the 243 ms delay and 20 cm offset resulted in the worst task performance across all synchrony conditions. Adding either a 243 ms delay to a 20 cm offset or a 20 cm offset to a 243 ms delay, significantly reduced task performance. Another thing worth noting is that a 98 ms delay or 12 cm offset on their own was not enough to significantly affect task performance compared to baseline. However, when combining them, it significantly affected all task performance metrics except time compared to baseline. Based on these findings, we suggest that even moderate offsets (12 cm), or low delays (98 ms), don’t affect task performance in tasks such as ours as long they are not combined. This is in line with [20], where they found the user’s ability to adapt to offsets in hand position remained as long as motion delays did not exceed 60 ms.

In a study by Yun et al. [52], they compared self-avatar animation conditions, two using inverse kinematics, and one using IMU motion capture. The motion capture had a 91 ms delay compared to the other conditions which had delays of 32-33 ms, it also had some issues with tracking drift. While motion capture was better in a copy pose task, one of the inverse kinematics conditions saw no significant difference from the motion capture in SoA and performed

better in collision avoidance manipulation tasks similar to ours. The authors suspected the added delay and drift in the motion capture condition might have contributed to this result. Based on our results, we think this could very well be the case since we found differences in SoA between 32 ms and 98 ms delay. Therefore, when measuring the SoA in self-avatars, we suggest that the conditions have similar delays unless the delay is accounted for in the study.

#### **4.2.4 PRQ1: Effect on the sense of body ownership**

In addition to the SoA, we measured the SoBO. Some studies have found that the SoBO is not affected by motion artifacts [28]. In [46], they found that it was affected by motion delay. We decided to investigate if our synchrony conditions affected the SoBO or not. However, it was not the main focus of the study, and since we only included two questionnaire statements for SoBO, the suggestions that follow from our results might not be as strong as those for SoA. We did expect our synchrony conditions, especially those with delay, to impact the SoBO negatively on the basis that temporal mismatch in visuotactile feedback has this effect [44]. We found that both motion delay and offset impacted SoBO negatively. Nevertheless, the impact on SoBO was not as strong as it was for SoA. Delay significantly impacted at 165 milliseconds, and offset at 20 centimeters. Furthermore, it should be noted that all mixed conditions negatively affected SoBO. Thus, smaller delays can affect it if combined with offset. In the worst synchrony condition, SoBO had almost completely deteriorated (scored on average 2.05). A majority (81.8 %) of participants reported in the post-questionnaire (PQ) that controlling the virtual upper body felt more like controlling an external tool when the synchrony was worse, suggesting body disownership. We also found a fairly strong positive correlation between SoA and SoBO.

#### **4.2.5 PRQ2: Correlation between immersive tendencies, SoA and SoBO**

We decided to investigate if we could find a relationship between participants' immersive tendencies and their SoA and SoBO, based on a finding by Waltemate et al. [46] which suggested that there is. We did not find a general correlation between either of SoA or SoBO and the average score in our sITQ questionnaire. However, we found weak to moderate correlations between SoA as well as SoBO in participants who reported that they were good at blocking out external distractions, and participants who reported that they could concentrate well on enjoyable activities. This suggests that there could be a small positive relationship between people who are better at immersing themselves in an activity and feeling that they're the agent of a virtual avatar.

#### **4.2.6 PRQ3: Synchrony and the attribution of agency**

How the SoA is experienced and is attributed to in our scenario could be explained in two layers. First, there is the feeling of agency over the self-avatar

arms that arises based on whether or not there is a match in visual and proprioceptive feedback when the user moves their arms. In terms of intention, action, and effect: the user intends to move the virtual arm to a certain position, their action is to move their real arm, and the (expected) effect is that the virtual arm moves accordingly. Discrepancies reduce the SoA. Secondly, there is the feeling of agency related to interactions with the virtual environment using the virtual arms, which could be argued is cognitive: being able to pick up and move an object to a goal. There is interplay between both layers since the user intention in the task becomes to move the virtual arms to move the virtual object. Both contribute to the overall SoA over the arms, and issues in either moving the virtual arms or being able to perform the task using the arms can reduce SoA. In particular, we’ve seen how delay impacts the SoA in self-avatars when there’s no environment interaction in [36], and how perceived task performance affects the SoA in [1]. We examined via post-questionnaire how the participants in our study attribute SoA over the virtual arms while performing the task when the synchrony felt better versus worse.

First of all, all participants reported that they experienced disturbances in the motion synchrony at some point in the experiment. This was expected since the worse synchrony conditions introduced large discrepancies between real and virtual motion. Participants reported that when they experienced that the synchrony was good, they paid more attention to the obstacles instead of their virtual arms (90.9 % of participants). In reality, when we perform a task using our body, we don’t usually think about how to move it. We believe that since users did not focus as much on how their virtual arms moved, they felt that the arms moved close to how they expected them to. This would allow them to focus on moving the object and completing the task, resulting in higher perceived task performance. Suggesting that both the feeling of agency over the virtual arms, and the feeling of agency over the interaction via their virtual arms, was high, resulting in a high overall SoA. Participants also felt more motivated when the synchrony was high (81.8 %), which might indicate that they felt more able to perform well in the task. High motivation could also be explained by the internal reward associated with a high SoA and how it promotes interaction via action frequency and action selection [25]. One can argue that when synchrony is high, moving the virtual arms might not be perceived as an effect or external event caused by the action of moving one’s real arms, but simply as the action itself. As discussed by Wen [48], this might also depend on whether there is a body ownership illusion over the arms.

On the other hand, participants reported that when they experienced that the synchrony was worse, they generally had a harder time focusing on the task (72.7 % of participants), had to use more mental effort (95.5 %), were more focused on the motion of the virtual arms than the obstacles in the task (90.9 %) and that the virtual arms felt more like external tools than arms that belonged to them (81.8 %). One participant reported that they were almost getting frustrated about the virtual arm motions when the synchrony was worse, and a few participants reported that the arms sometimes felt like someone else’s. Since participants had to focus more on the motion of their virtual arms and

expend more mental effort, this probably resulted in feeling a lower agency over the virtual arms. Worse control over the virtual arms also meant worse task performance, which translates to less feeling of agency over the interaction when using their virtual arms. Making the negative effect two-fold, and in total resulting in a lower overall SoA.

We believe that the combination of controlling the self-avatar’s virtual arms and using them to perform a task in the virtual environment allowed us to find that even small changes in synchrony negatively impact the overall SoA. In comparison, tasks when the user simply moves their self-avatar and observes might not be as susceptible since there is no external goal of the movement that could be affected by low synchrony. Tasks that don’t require the user to be aware of their self-avatar motion might not be susceptible at all. Therefore, to promote interaction and engagement in VR application where users have to interact with the virtual environment using a self-avatar, the synchrony of the self-avatar is critical since it maintains the SoA.

On a separate note regarding the attribution of SoA, we think an intriguing research topic would be to investigate how physics-based interactions with self-avatars affect the SoA. For instance, if an object has simulated weight, grabbing that object may cause the virtual arm pose to be affected. The user might expect a clump of metal to be heavy and pull their virtual arm down (via gravity), but they might also expect the virtual arm to match their physical arm pose. It would be interesting to examine if simulated physics are important for improving the SoA, or if it is more important that the virtual arm pose is correct, or if there is a middle ground. One could even incorporate real-world objects with physical weight in such a study.

#### **4.2.7 Limitations**

While this study provides valuable insights into the impact of disturbed motion synchrony in self-avatars, it is important to recognize certain limitations in terms of the methodology and the generalizability of the findings.

Regarding generalizability, the task was crafted in such a way that being aware of one’s arm motion is intrinsic to performing the task well. In general manipulation tasks performed in virtual environments, this kind of arm motion awareness is typically not required. In such tasks, disruptions in synchrony might not have the same impact on the SoA. Still, since our task serves as a kind of worst-case, our findings provide thresholds for which to keep delay and offset above in general to ensure low impact.

There are a few limitations in the methodology. One of them is that we did not measure the time between when a task was spawned and when the participant grabbed the pickup. During this time, the self-avatar was visible, which means that the participants could observe their motion without focusing on the task. Therefore, this duration is an unknown variable that differed among participants and might have impacted their reported SoA and SoBO. For example, participants who waited longer to grab the pickup might have experienced a lower SoA since they had the opportunity to observe their arm

motion for a longer period. However, based on our observations, all participants spent at most a couple of seconds waiting to grab the pickup in each task. In the post-questionnaire, 45.4 % of participants responded that the first motion of moving their virtual arm toward the pickup object and grabbing it was what made it most apparent that the arms were less synchronous. However, we think the wording of the corresponding statement in the questionnaire creates a biased response. Since, in the worse synchrony conditions, participants would immediately recognize differences in the arm motion as soon as they started moving their arms. Instead of letting participants decide when to start the task by grabbing the pickup, there could be a hidden timer that determines when the task starts. This way, all participants would have the same period of time before each task starts. Alternatively, tasks could start immediately after one another.

A limitation of the self-avatar implementation is that we estimated the arm pose and the body forward direction. While the layout of each task was designed with the elbow bend direction in mind (such that there would be no necessary collisions for completing any task); three participants reported that they noticed that the elbows were not accurate (all with high VR experience). In our second pilot, we explicitly asked each participant if they noticed this limitation, and no one did. Therefore, we would argue that it is not entirely reasonable to extrapolate that a lot more than three participants noticed this limitation. However, for any participant who did notice it, it can have had an impact on their reported SoA and SoBO. For example, during the worse synchrony conditions, this mismatch in elbow position might have become more evident since participants paid more attention to their virtual arms then. Which, presumably would influence the reported SoA. Similar reasoning could be used for the body forward estimation, however, only one participant with high VR experience commented about it. To remove the need for elbow and body forward estimation while keeping the baseline delay low, HTC Vive trackers with base stations could be utilized.

It could be argued that measuring the SoA via questionnaire is a limitation as well since this type of measurement is subject to individual criteria. It is possible to measure the SoA implicitly and three mechanisms for doing so are described in [49]. Incorporating such measurements in self-avatar studies about the SoA could yield new insights.

#### **4.2.8 A note on sustainability and ethics**

While it is not entirely clear how to put this study directly in the context of sustainability and ethics, at least certain points of discussion can be had.

From a social sustainability perspective, requirements on synchrony in self-avatars can have implications for social interactions in long-range communication and collaboration. With the internet and social media, communicating and collaborating with people in different parts of the world has never been easier. Some companies, for example Meta, are betting heavily on VR (maybe more specifically MR - mixed reality) being the foundation for the next large social

platform. Such a platform could bring people closer together by enhancing co-presence in virtual environments, where having a self-avatar would enable users to express themselves through body language in social interactions. Feeling a sense of agency over your self-avatar can be important in those situations, which means that requirements on synchrony could be informative. Similar arguments about sustainability could be made from an economic or ecological perspective. With hybrid work conditions already being a reality, self-avatars that match the user’s body motion could enhance social interaction in digital meetings, possibly improving long-range collaborative ability and productivity. If the quality of digital meetings in VR keep improving, they could reduce the need for long-range travel, possibly reducing carbon dioxide output (provided server halls and production of hardware are efficient enough). Toothman and Neff [43] investigate the impact of motion artifacts in self-avatars on social aspects in closer detail.

On a more specific note, in terms of ethics, the self-avatar in our user study did not accomodate for participants’ body shape or size. Since the avatar had a certain body type, participants who had a similar body type might have felt more comfortable than those who did not. In studies with more resources, several avatars are sometimes implemented to reflect different skin colors, gender, and body type. Regarding the experiment procedure, we made sure participants were well situated in the room and we never had to intervene to prevent any collision with physical walls or furniture during the experiments. However, one of the rooms was somewhat small and there was a slight risk that participants could collide with the physical environment. Additionally, there is always the risk of eye strain or simulator sickness when in VR for extended periods of time.

The broader ethical aspects of the work are harder to discuss. We don’t think it is reasonable to suggest that the findings of this work could be used in harmful ways. A very far-fetched idea could be that this knowledge of how to disrupt the sense of agency in self-avatars could be used by bad actors to make informed decisions about how to negatively impact user activities in VR. If we in the future spend a lot of time inside of virtual environments, this could possibly become a problem, but I reason that is still unlikely.

## 5 Conclusion

We conducted a user study to investigate the impact of disturbed synchrony via motion delay and offset, between user’s real and virtual arms (using VR) on their sense of agency (SoA) and task performance, but also to a lesser extent their sense of body ownership (SoBO). Our results suggest that both delay, offset, and combinations of them affect SoA, SoBO, and task performance. We found a new lower threshold for when delay negatively affects the SoA in tasks where the user has to be highly aware of their arm motion and interact with the virtual environment, at 98 milliseconds. Arm motion delay had a larger impact on SoA than arm offset. However, the impact of offset was significant. Surprisingly, we found that SoA was impacted by smaller disruptions in synchrony than task performance, we speculate because of our task design. The SoBO was not found to be as sensitive to disturbed synchrony as the SoA but was still negatively affected. Combining both motion delay and offset had the largest impact on our metrics. Small delays with moderate offsets had a significant effect on task performance, even when those delays and offsets applied individually did not. We also found that when synchrony is disturbed, users have to expend more mental effort to focus on the self-avatar motion and focus less on the task at hand.

Even though the study had some limitations, these findings suggest that even small disruptions in motion synchrony between users and their self-avatars can negatively affect their subjective experience and performance in VR applications, especially when the user has to be highly aware of their body motion and use the self-avatar to perform tasks in the virtual environment. Therefore, they have implications for developing immersive VR experiences. Additionally, our findings could aid by informing in the development of motion tracking or motion reconstruction for self-avatars. Separate from self-avatars in VR, we think this knowledge of how delay and offset affect the SoA and task performance in manipulation tasks could aid in other real-world applications such as when tele-operating robotic arms. Especially if they are far away, for instance on the other side of the globe, or eventually on the moon, where motion delays could be super second long, and offsets are subject to the design of the robotics.

In the future, we think it could be interesting to investigate even lower motion delays than 98 milliseconds since that still harmed the SoA and there could be lower thresholds in similar tasks. Examining similar tasks but for the full body would also be interesting. Especially to see how offsets could impact other parts of the body, for instance, the feet. On a separate note, we think it would be very informative to examine the impact of physics-based interactions with self-avatars on the SoA, where for instance simulated weight of objects affects the motion of the virtual arms. Finally, incorporating implicit measurements of the SoA in self-avatar studies could yield new valuable insights.

## References

- [1] Kei Aoyagi, Wen Wen, Qi An, Shunsuke Hamasaki, Hiroshi Yamakawa, Yusuke Tamura, Atsushi Yamashita, and Hajime Asama. 2021. Modified sensory feedback enhances the sense of agency during continuous body movements in virtual reality. *Scientific Reports* 11, 1 (Jan. 2021), 2553. <https://doi.org/10.1038/s41598-021-82154-y>.
- [2] Andreas Aristidou, Joan Lasenby, Yiorgos Chrysanthou, and Ariel Shamir. 2018. Inverse Kinematics Techniques in Computer Graphics: A Survey. *Computer Graphics Forum* 37, 6 (2018), 35–58. <https://doi.org/10.1111/cgf.13310>.
- [3] Frank Biocca. 1992. Virtual Reality Technology: A Tutorial. *Journal of Communication* 42, 4 (1992), 23–72. <https://doi.org/10.1111/j.1460-2466.1992.tb00811.x>.
- [4] Sarah-J Blakemore, Daniel M Wolpert, and Chris D Frith. 1998. Central cancellation of self-produced tickle sensation. *Nature neuroscience* 1, 7 (1998), 635–640. <https://doi.org/10.1038/2870>
- [5] Matthew Botvinick and Jonathan Cohen. 1998. Rubber hands ‘feel’ touch that eyes see. *Nature* 391, 6669 (1998), 756–756. <https://doi.org/10.1038/35784>
- [6] Sidney Bovet, Henrique Galvan Debarba, Bruno Herbelin, Eray Molla, and Ronan Boulic. 2018. The Critical Role of Self-Contact for Embodiment in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 24, 4 (April 2018), 1428–1436. <https://doi.org/10.1109/TVCG.2018.2794658>.
- [7] Polona Caserman, Philipp Achenbach, and Stefan Göbel. 2019. Analysis of Inverse Kinematics Solutions for Full-Body Reconstruction in Virtual Reality. In *2019 IEEE 7th International Conference on Serious Games and Applications for Health (SeGAH)*. 1–8. <https://doi.org/10.1109/SeGAH.2019.8882429>.
- [8] Polona Caserman, Augusto Garcia-Agundez, and Stefan Gobel. 2020. A Survey of Full-Body Motion Reconstruction in Immersive Virtual Reality Applications. *IEEE Transactions on Visualization and Computer Graphics* 26, 10 (Oct. 2020), 3089–3108. <https://doi.org/10.1109/TVCG.2019.2912607>
- [9] Sebastian Cmentowski and Jens Krüger. 2021. Effects of Task Type and Wall Appearance on Collision Behavior in Virtual Environments. In *2021 IEEE Conference on Games (CoG)*. 1–8. <https://doi.org/10.1109/CoG52621.2021.9619039>

- [10] Ralph B. D’Agostino and Egon Sharpe Pearson. 1973. Tests for departure from normality. Empirical results for the distributions of  $b_2$  and  $\sqrt{b_1}$ . *Biometrika* 60 (1973), 613–622. <https://doi.org/10.1093/BIOMET/60.3.613>
- [11] Henrique Galvan Debarba, Sylvain Chagué, and Caecilia Charbonnier. 2022. On the Plausibility of Virtual Body Animation Features in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 28, 4 (April 2022), 1880–1893. <https://doi.org/10.1109/TVCG.2020.3025175>.
- [12] Yuming Du, Robin Kips, Albert Pumarola, Sebastian Starke, Ali Thabet, and Artsiom Sanakoyeu. 2023. Avatars Grow Legs: Generating Smooth Human Motion from Sparse Tracking Inputs with Diffusion Model. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 481–490. <https://doi.org/10.1109/CVPR52729.2023.00054>
- [13] James Coleman Eubanks, Alec G. Moore, Paul A. Fishwick, and Ryan P. McMahan. 2020. The Effects of Body Tracking Fidelity on Embodiment of an Inverse-Kinematic Avatar for Male Participants. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 54–63. <https://doi.org/10.1109/ISMAR50242.2020.00025>.
- [14] Nicolas Franck, Chloé Farrer, Nicolas Georgieff, Michel Marie-Cardine, Jean Daléry, Thierry d’Amato, and Marc Jeannerod. 2001. Defective Recognition of One’s Own Actions in Patients With Schizophrenia. *American Journal of Psychiatry* 158, 3 (March 2001), 454–459. <https://doi.org/10.1176/appi.ajp.158.3.454>.
- [15] Rebecca Fribourg, Ferran Argelaguet, Anatole Lécuyer, and Ludovic Hoyet. 2020. Avatar and Sense of Embodiment: Studying the Relative Preference Between Appearance, Control and Point of View. *IEEE Transactions on Visualization and Computer Graphics* 26, 5 (May 2020), 2062–2072. <https://doi.org/10.1109/TVCG.2020.2973077>.
- [16] Chris D Frith, Sarah-Jayne Blakemore, and Daniel M Wolpert. 2000. Explaining the symptoms of schizophrenia: Abnormalities in the awareness of action. *Brain Research Reviews* 31, 2 (2000), 357–363. [https://doi.org/10.1016/S0165-0173\(99\)00052-1](https://doi.org/10.1016/S0165-0173(99)00052-1)
- [17] Francesca Garbarini, Angela Mastropasqua, Monica Sigauo, Marco Rabuffetti, Alessandro Piedimonte, Lorenzo Pia, and Paola Rocca. 2016. Abnormal sense of agency in patients with schizophrenia: Evidence from bi-manual coupling paradigm. *Frontiers in Behavioral Neuroscience* 10 (2016), 43. <https://doi.org/10.3389/fnbeh.2016.00043>
- [18] Joris Groen and Peter J. Werkhoven. 1998. Visuomotor Adaptation to Virtual Hand Position in Interactive Virtual Environments. *Presence* 7, 5 (Oct. 1998), 429–446. <https://doi.org/10.1162/105474698565839>.

- [19] Chenyu Gu, Weicon Lin, Xinyi He, Lei Zhang, and Mingming Zhang. 2023. IMU-based motion capture system for rehabilitation applications: A systematic review. *Biomimetic Intelligence and Robotics* 3, 2 (2023), 100097. <https://doi.org/10.1016/j.birob.2023.100097>
- [20] Richard Held and Nathaniel Durlach. 1993. Telepresence, time delay and adaptation. In *Pictorial Communication in Virtual and Real Environments (2nd Ed.)*. Taylor & Francis, Inc., USA, 232–246.
- [21] Richard Held, Aglaia Efstathiou, and Martha Greene. 1966. Adaption to displaced and delayed visual feedback from the hand. *Journal of Experimental Psychology* 72 (12 1966), 887–891. <https://doi.org/10.1037/h0023868>
- [22] Valentin Holzwarth, Joy Gisler, Christian Hirt, and Andreas Kunz. 2021. Comparing the Accuracy and Precision of SteamVR Tracking 2.0 and Oculus Quest 2 in a Room Scale Setup. In *Proceedings of the 2021 5th International Conference on Virtual and Augmented Reality Simulations (ICVARS '21)*. Association for Computing Machinery, New York, NY, USA, 42–46. <https://doi.org/10.1145/3463914.3463921>
- [23] Emma E Howard, S Gareth Edwards, and Andrew P Bayliss. 2016. Physical and mental effort disrupts the implicit sense of agency. *Cognition* 157 (2016), 114–125. <https://doi.org/10.1016/j.cognition.2016.08.018>
- [24] Andreas Kalckert and H. Henrik Ehrsson. 2012. Moving a Rubber Hand that Feels Like Your Own: A Dissociation of Ownership and Agency. *Frontiers in Human Neuroscience* 6 (March 2012). <https://doi.org/10.3389/fnhum.2012.00040> .
- [25] Noam Karsh and Baruch Eitam. 2015. I control therefore I do: Judgments of agency influence action selection. *Cognition* 138 (2015), 122–131. <https://doi.org/10.1016/j.cognition.2015.02.002>
- [26] Samantha Keenaghan, Lucy Bowles, Georgina Crawford, Simon Thurlbeck, Robert W. Kentridge, and Dorothy Cowie. 2020. My body until proven otherwise: Exploring the time course of the full body illusion. *Consciousness and Cognition* 78 (Feb. 2020), 102882. <https://doi.org/10.1016/j.concog.2020.102882>
- [27] Konstantina Kilteni, Raphaella Groten, and Mel Slater. 2012. The Sense of Embodiment in Virtual Reality. *Presence* 21, 4 (Nov. 2012), 373–387. [https://doi.org/10.1162/PRES\\_a\\_00124](https://doi.org/10.1162/PRES_a_00124) .
- [28] Alexandros Koiliias, Christos Mousas, and Christos-Nikolaos Anagnostopoulos. 2019. The Effects of Motion Artifacts on Self-Avatar Agency. *Informatics* 6, 2 (June 2019), 18. <https://doi.org/10.3390/informatics6020018> .

- [29] Chris Miall and Daniel Wolpert. 1996. Forward Models for Physiological Motor Control. *Neural Networks* 9, 8 (1996), 1265–1279. [https://doi.org/10.1016/S0893-6080\(96\)00035-4](https://doi.org/10.1016/S0893-6080(96)00035-4).
- [30] Betty J Mohler, Sarah H Creem-Regehr, William B Thompson, and Heinrich H Bühlhoff. 2010. The Effect of Viewing a Self-Avatar on Distance Judgments in an HMD-Based Virtual Environment. *Presence: Teleoperators and Virtual Environments* 19, 3 (June 2010), 230–242. <https://doi.org/10.1162/pres.19.3.230>
- [31] James W Moore and Paul C Fletcher. 2012. Sense of agency in health and disease: a review of cue integration approaches. *Consciousness and cognition* 21, 1 (2012), 59–68. <https://doi.org/10.1016/j.concog.2011.08.010>
- [32] Aske Mottelson, Andreea Muresan, Kasper Hornbæk, and Guido Makransky. 2023. A systematic review and meta-analysis of the effectiveness of body ownership illusions in virtual reality. *ACM Transactions on Computer-Human Interaction* 30, 5 (2023), 1–42. <https://doi.org/10.1145/3590767>
- [33] Ye Pan and Anthony Steed. 2019. Avatar Type Affects Performance of Cognitive Tasks in Virtual Reality. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology* (Parramatta, NSW, Australia) (VRST '19). Association for Computing Machinery, New York, NY, USA, Article 6, 4 pages. <https://doi.org/10.1145/3359996.3364270>
- [34] Mathias Parger, Joerg H. Mueller, Dieter Schmalstieg, and Markus Steinberger. 2018. Human upper-body inverse kinematics for increased embodiment in consumer-grade virtual reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology* (VRST '18). Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/3281505.3281529>
- [35] Thibault Porssut, Olaf Blanke, Bruno Herbelin, and Ronan Boulic. 2022. Reaching articular limits can negatively impact embodiment in virtual reality. *PLOS ONE* 17, 3 (March 2022), e0255554. <https://doi.org/10.1371/journal.pone.0255554>.
- [36] Daniel Roth and Marc Erich Latoschik. 2020. Construction of the Virtual Embodiment Questionnaire (VEQ). *IEEE Transactions on Visualization and Computer Graphics* 26, 12 (Dec. 2020), 3546–3556. <https://doi.org/10.1109/TVCG.2020.3023603>.
- [37] Maria V Sanchez-Vives, Bernhard Spanlang, Antonio Frisoli, Massimo Bergamasco, and Mel Slater. 2010. Virtual hand illusion induced by visuomotor correlations. *PloS one* 5, 4 (2010), e10381. <https://doi.org/10.1371/journal.pone.0010381>

- [38] Frauke Schultze-Lutter. 2009. Subjective symptoms of schizophrenia in research and the clinic: the basic symptom concept. *Schizophrenia bulletin* 35, 1 (2009), 5–8. <https://doi.org/10.1093/schbul/sbn139>
- [39] Mel Slater, Bernhard Spanlang, and David Corominas. 2010. Simulating virtual environments within virtual environments as the basis for a psychophysics of presence. *ACM Transactions on Graphics* 29, 4 (July 2010), 92:1–92:9. <https://doi.org/10.1145/1778765.1778829>
- [40] Marco Sperduti, Pauline Delaveau, Philippe Fossati, and Jaqueline Nadel. 2011. Different brain structures related to self-and external-agency attribution: a brief review and meta-analysis. *Brain Structure and Function* 216 (2011), 151–157. <https://doi.org/10.1007/s00429-010-0298-1>
- [41] Anthony Steed, Ye Pan, Fiona Zisch, and William Steptoe. 2016. The impact of a self-avatar on cognitive load in immersive virtual reality. In *2016 IEEE Virtual Reality (VR)*. 67–76. <https://doi.org/10.1109/VR.2016.7504689>
- [42] Matthis Synofzik, Gottfried Vosgerau, and Albert Newen. 2008. Beyond the comparator model: A multifactorial two-step account of agency. *Consciousness and Cognition* 17, 1 (2008), 219–239. <https://doi.org/10.1016/j.concog.2007.03.010>
- [43] Nicholas Toothman and Michael Neff. 2019. The Impact of Avatar Tracking Errors on User Experience in VR. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 756–766. <https://doi.org/10.1109/VR.2019.8798108> .
- [44] Manos Tsakiris and Patrick Haggard. 2005. The rubber hand illusion revisited: visuotactile integration and self-attribution. *Journal of experimental psychology: Human perception and performance* 31, 1 (2005), 80. <https://doi.org/10.1037/0096-1523.31.1.80>
- [45] John C Tuthill and Eiman Azim. 2018. Proprioception. *Current Biology* 28, 5 (2018), R194–R203. <https://doi.org/10.1016/j.cub.2018.01.064>
- [46] Thomas Waltemate, Irene Senna, Felix Hülsmann, Marieke Rohde, Stefan Kopp, Marc Ernst, and Mario Botsch. 2016. The impact of latency on perceptual judgments and motor performance in closed-loop interaction in virtual reality. In *Proceedings of the 22nd ACM Conference on Virtual Reality Software and Technology (VRST '16)*. Association for Computing Machinery, New York, NY, USA, 27–35. <https://doi.org/10.1145/2993369.2993381>
- [47] Matthew Warburton, Mark Mon-Williams, Faisal Mushtaq, and J. Ryan Morehead. 2023. Measuring motion-to-photon latency for sensorimotor experiments with virtual reality systems. *Behavior Research Methods* 55, 7 (Oct. 2023), 3658–3678. <https://doi.org/10.3758/s13428-022-01983-5>

- [48] Wen Wen. 2019. Does delay in feedback diminish sense of agency? A review. *Consciousness and Cognition* 73 (Aug. 2019), 102759. <https://doi.org/10.1016/j.concog.2019.05.007>
- [49] Wen Wen and Hiroshi Imamizu. 2022. The sense of agency in perception, behaviour and human-machine interactions. *Nature Reviews Psychology* 1, 4 (April 2022), 211–222. <https://doi.org/10.1038/s44159-022-00030-6>.
- [50] Bob G. Witmer and Michael J. Singer. 1998. Measuring Presence in Virtual Environments: A Presence Questionnaire. *Presence: Teleoperators and Virtual Environments* 7, 3 (June 1998), 225–240. <https://doi.org/10.1162/105474698565686>
- [51] Daniel Wolpert, Zoubin Ghahramani, and Michael Jordan. 1995. An Internal Model for Sensorimotor Integration. *Science (New York, N.Y.)* 269 (10 1995), 1880–2. <https://doi.org/10.1126/science.7569931>
- [52] Haoran Yun, Jose Luis Ponton, Carlos Andujar, and Nuria Pelechano. 2023. Animation Fidelity in Self-Avatars: Impact on User Performance and Sense of Agency. In *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. 286–296. <https://doi.org/10.1109/VR55154.2023.00044>.

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